

Last Time Practice Series

WATE-2023

Digital Electronics Part-2

By Sanjay Rath Sir



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- ❖ Teaching Experience: 21+ Years
- ❖ Achievements:
 - AIR-16 (ESE 2005), AIR-21 (ESE 2008)
 - Mentored More Than 1,00,000+ Students
- ❖ Area of Expertise: Network Theory, Digital Circuits, Control Systems, EDC, Analog Circuits, Advanced Electronics

UPSC



SANJAY RATHI

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$$\begin{array}{r} 14 \\ 22 \end{array} \quad \begin{array}{r} 6 \times 1 \\ 8 \times 2 = 16 \end{array} \quad \begin{array}{r} 6 \\ 22 \end{array}$$

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$$\overline{I_0} \overline{I_1} = D_0 \quad D_2 = I_1 \overline{I_0}$$

$$Y_1 = D_2 + D_0 = I_1 \overline{I_0} + \overline{I_1} \overline{I_0}$$

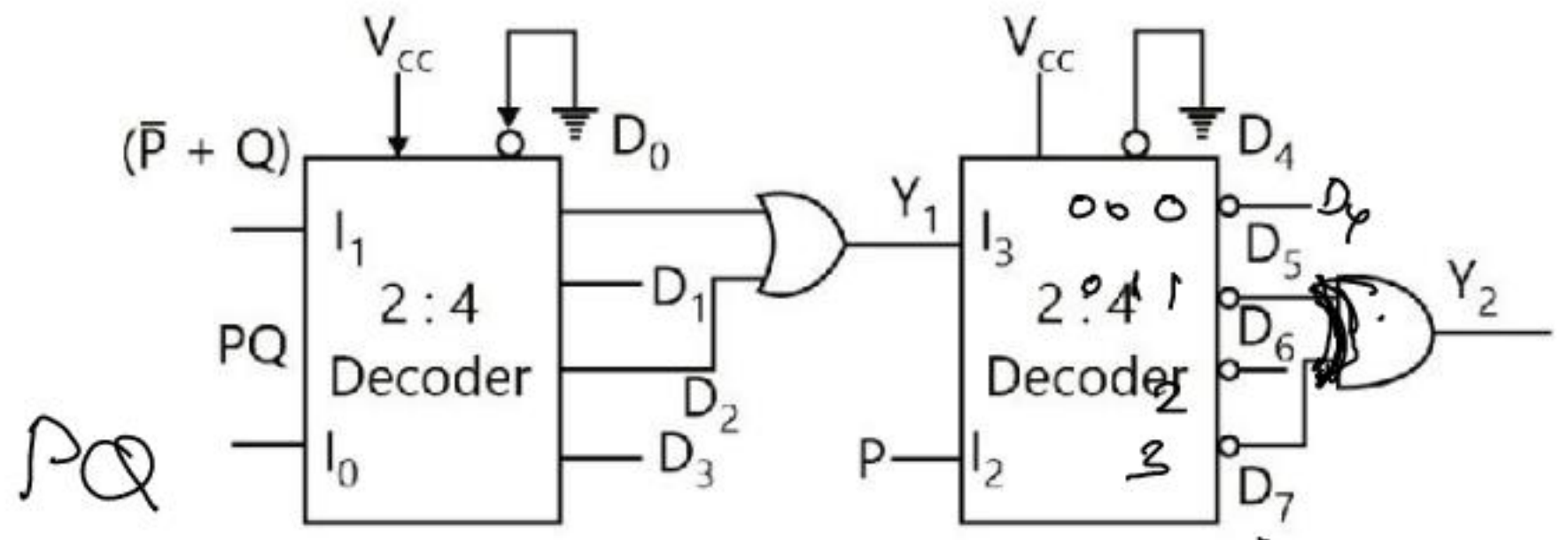
$$\overline{I_0} [I_1 + \overline{I_1}] = \overline{I_0}$$

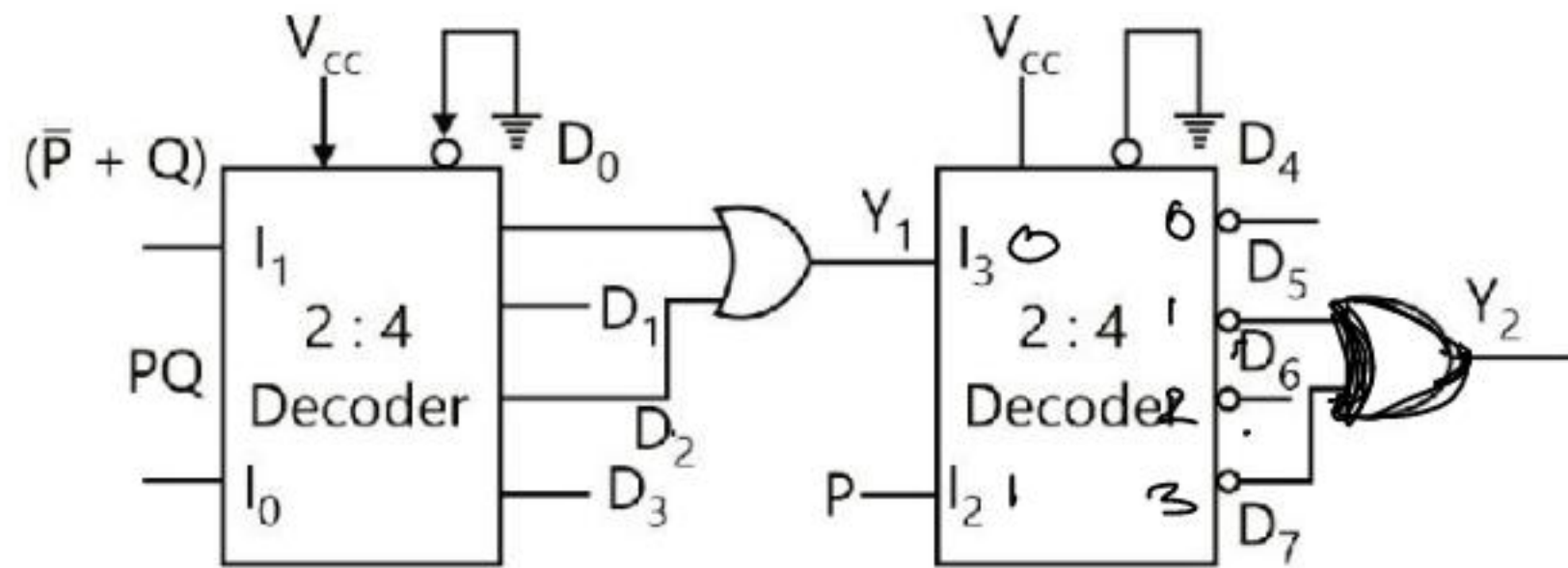
$$Y_1 = \overline{PQ}$$

$$\begin{aligned} Y_2 &= \overline{D_5} + \overline{D_6} \\ &= \overline{I_3 I_2} + \overline{I_3 I_2} \\ &= \overline{I_2} [\overline{I_3} + I_3] \\ &= \overline{I_2} \end{aligned}$$

Q. From the given decoder circuit shown below, find output Y1 and Y2

- A. Y1 is NAND of P, Q and Y2 = $\overline{P}$ (c) None
- B. Y1 is NOR of P, Q and Y2 = $\overline{P}$
- C. Y1 is NAND of P, Q and Y2 = P
- D. Y1 is NOR of P & Q and Y2 = P





$$Y_2 = \overline{D_5} + \overline{D_6}$$

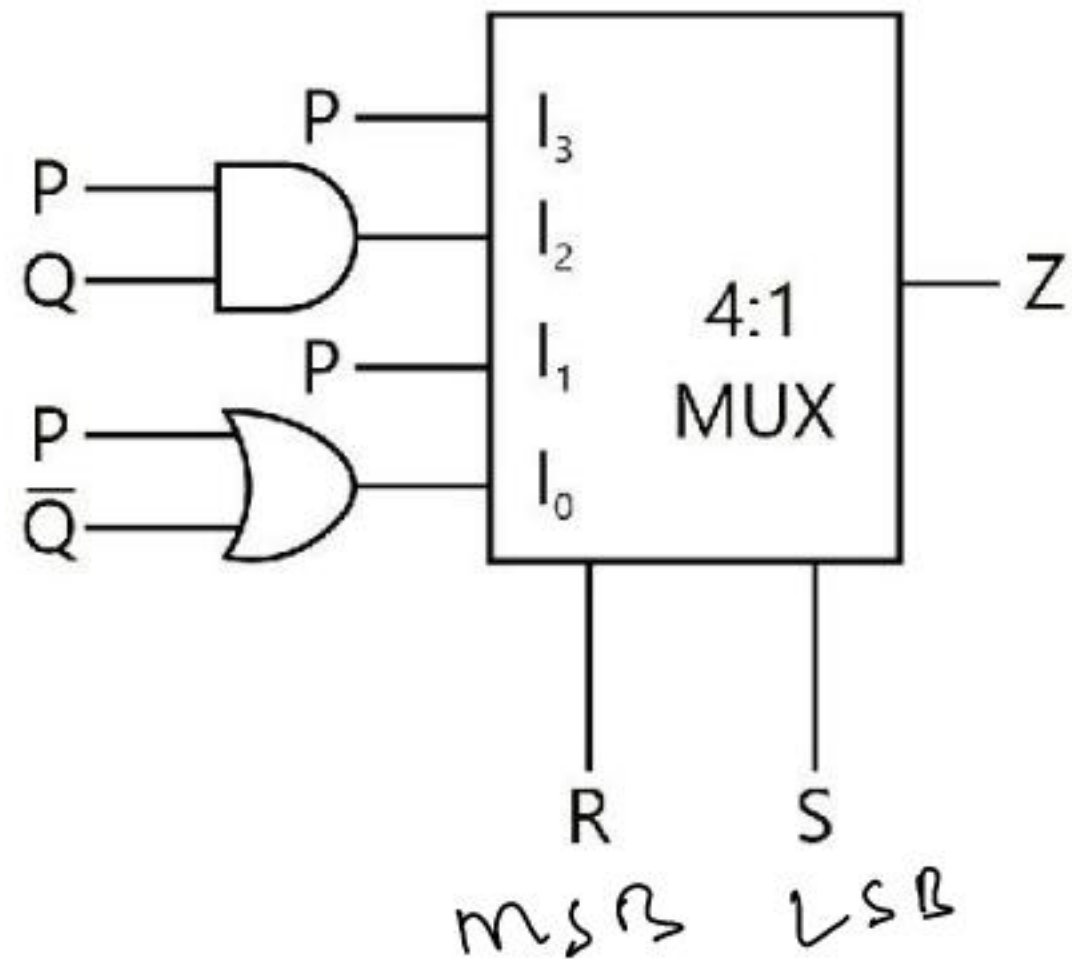
$$= \overline{\overline{I_3} I_2} + \overline{I_3 \overline{I_2}}$$

$$= (\overline{\overline{I_3} + \overline{I_2}}) + (\overline{\overline{I_3} + \overline{I_2}})$$

$$= \overline{I_3} + \overline{\overline{I_2}} + \overline{\overline{I_3}} + \overline{I_2}$$

$$= 1 + \overline{I_2}$$

$$P = 1$$



Q. For the circuit shown in the following figure, $I_0 - I_3$ are inputs to the 4:1 multiplexer. R (MSB) and S are control bits,

The output Z can be represented by

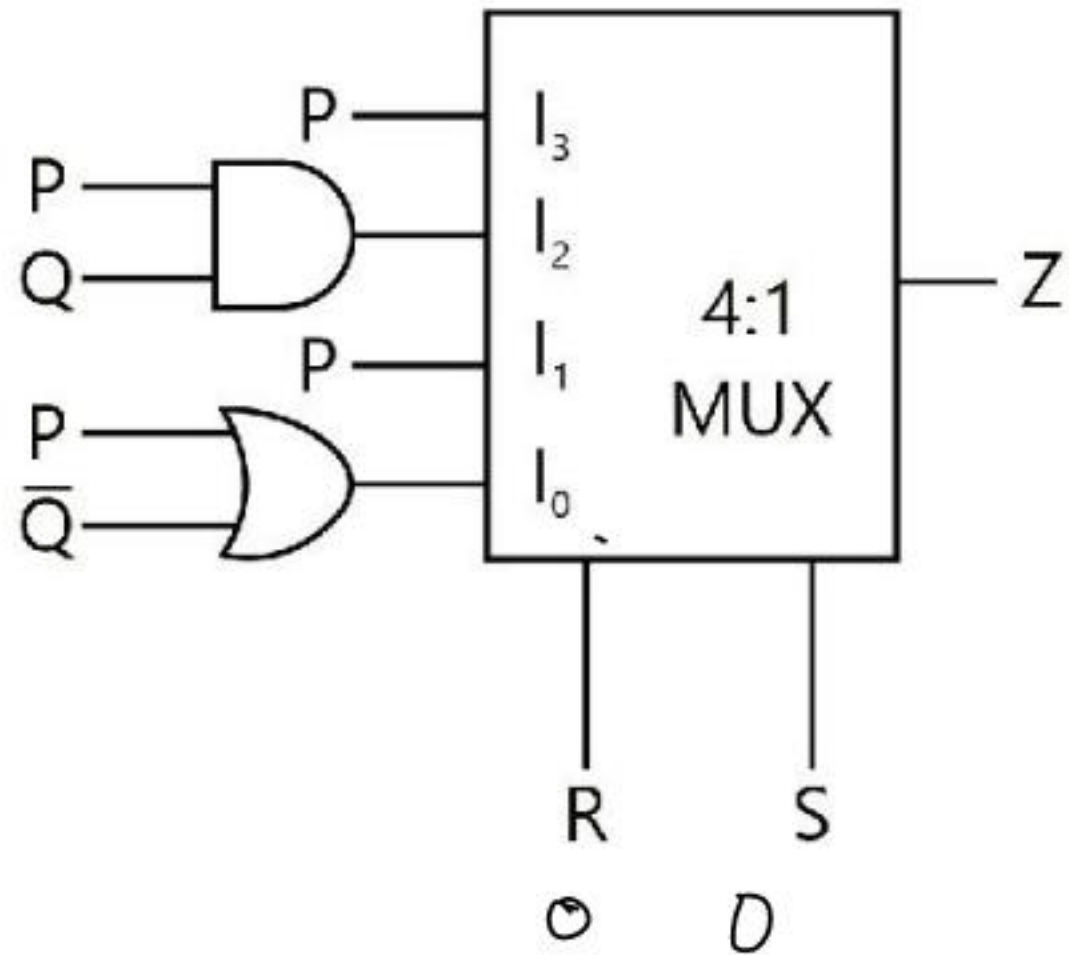
☒ A. $\underline{PQ} + \underline{P\bar{Q}S} + \underline{\bar{Q}R\bar{S}}$

B. $\underline{P\bar{Q}} + \underline{PQR} + \underline{\bar{P}\bar{Q}\bar{S}}$

C. $\underline{P\bar{Q}R} + \underline{\bar{P}QR} + \underline{PQRS} + \underline{\bar{Q}R\bar{S}}$

D. $\underline{PQ\bar{R}} + \underline{PQR\bar{S}} + \underline{P\bar{Q}R\bar{S}} + \underline{\bar{Q}R\bar{S}}$

[GATE-2008, 2 MARKS]



Truth table for the 4:1 MUX circuit:

	$\overline{P}\overline{Q}$	$\overline{P}Q$	$P\overline{Q}$	PQ
I_0 $\overline{R}\overline{S}$	1		1	1
I_1 $\overline{R}S$			1	
I_2 $R\overline{S}$			1	
I_3 RS				1

Handwritten notes: $I_0 = P + \overline{Q}$ and $I_2 = P\overline{Q}$.

$$I_0 = P + \overline{Q}$$

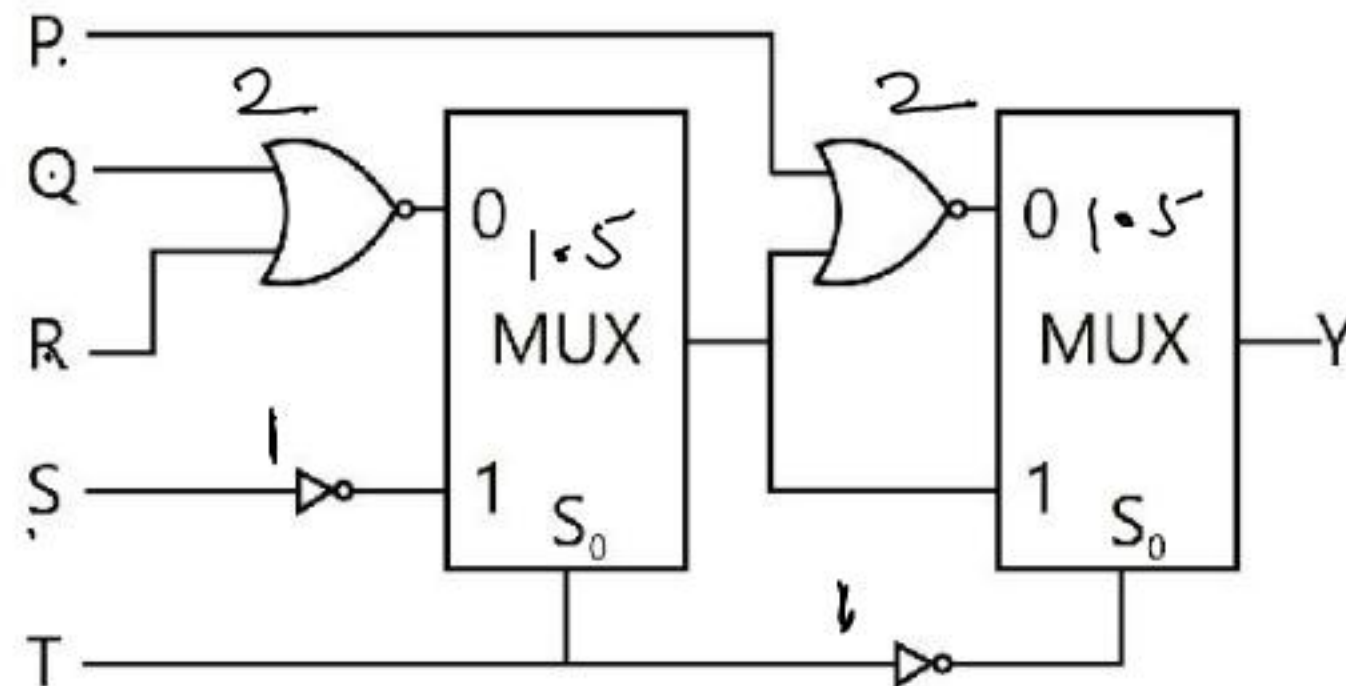
$$I_2 = P\overline{Q}$$

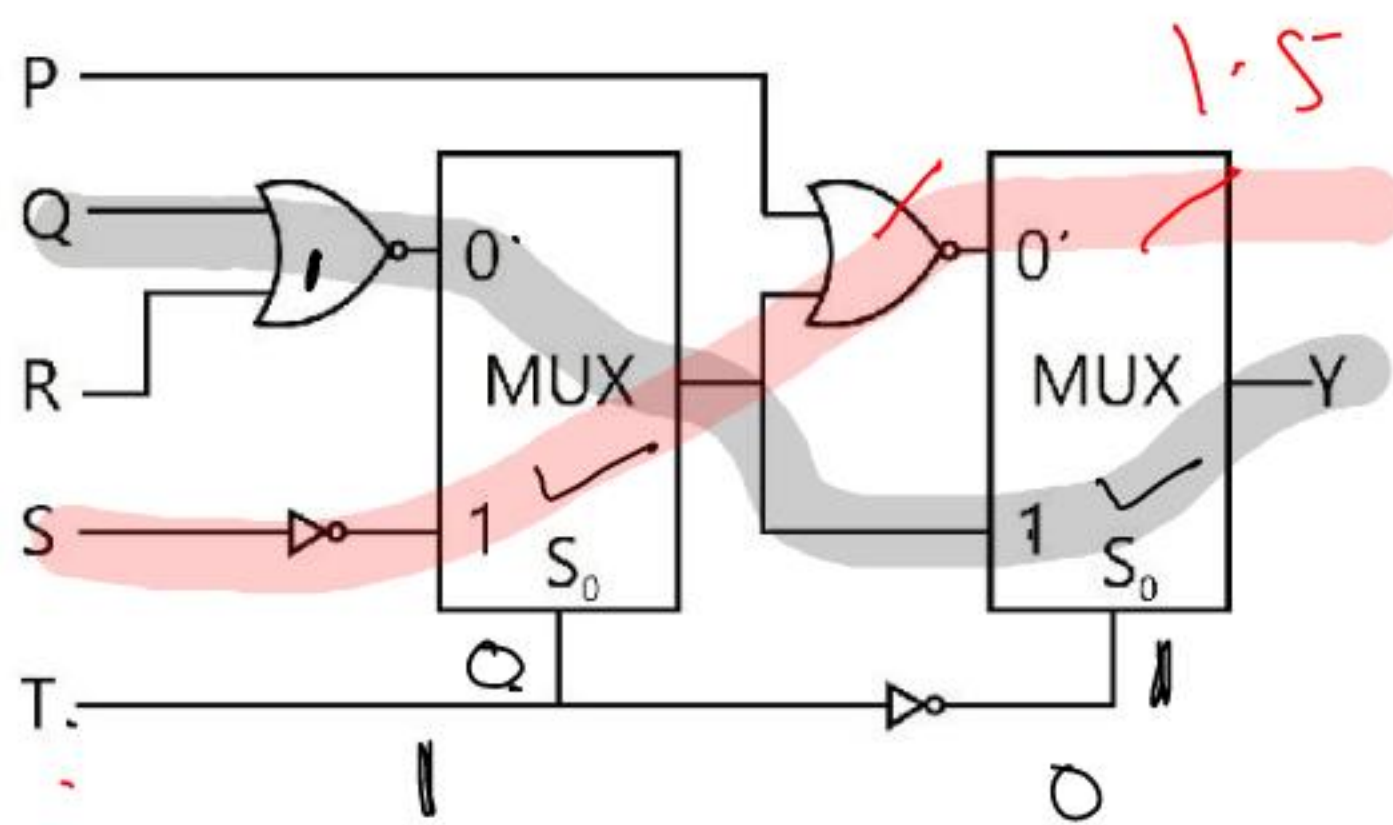
$$\begin{aligned}
 Y &= \bar{S}_1 \bar{S}_0 \bar{I}_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 \bar{I}_2 + S_1 S_0 I_3 \\
 &= \bar{R} \bar{S} [P + \bar{Q}] + \bar{R} S P + R \bar{S} P Q + R S P \\
 &= \bar{R} \bar{S} P + \bar{R} \bar{S} \bar{Q} + \bar{R} S P + R \bar{S} P Q + R S P
 \end{aligned}$$

	$\bar{p}\bar{q}$	$\bar{p}q$	pq	$p\bar{q}$
$\bar{R}\bar{S}$	1		1	1
$\bar{R}S$			1	1
RS			1	1
$R\bar{S}$			1	

Q. For the circuit shown in the figure, the delays of NOR gates, multiplexers and inverters are 2 ns, 1.5 ns and 1 ns, respectively. If all the inputs P, Q, R, S and T are applied at the same time instant, the maximum propagation delay (in ns) of the circuit is _____.

[GATE-2016, 2 MARKS]





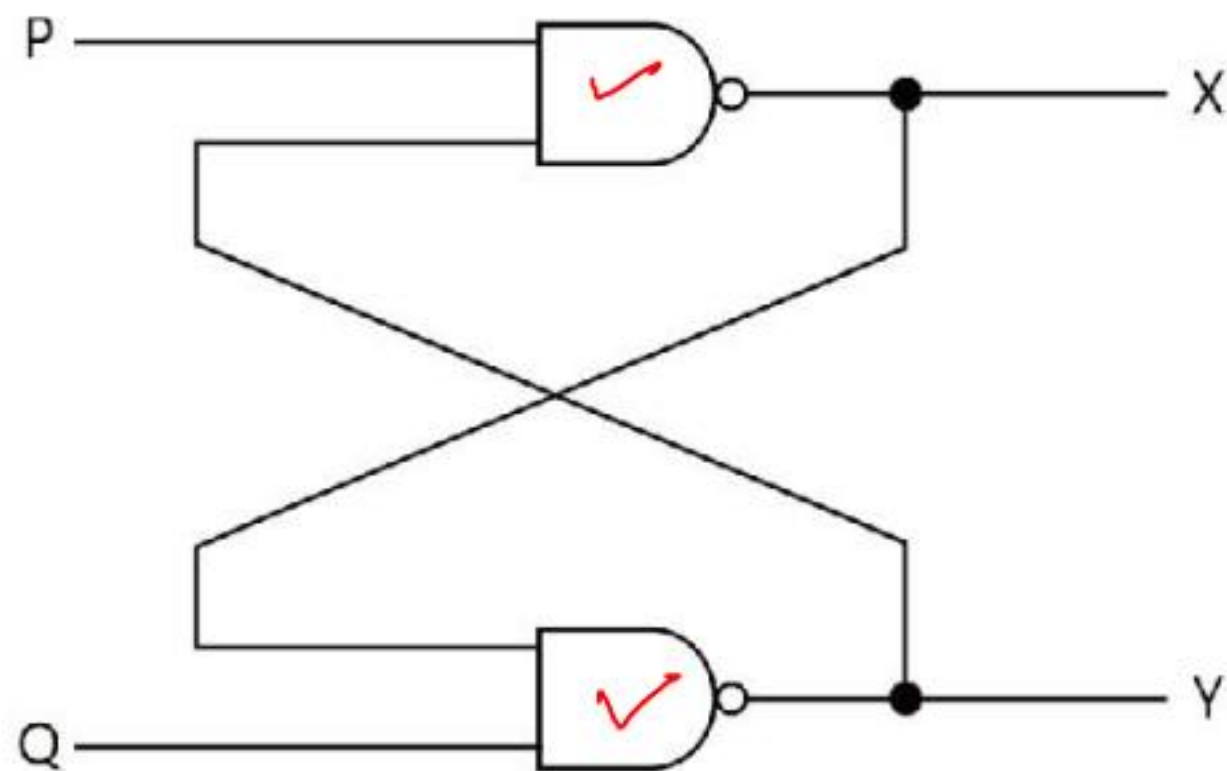
$$\underline{T = 0}$$

$$\tau_D = 2 + 1.5 + 1.5 = \underline{\underline{5 \text{ nsec}}}$$

$$\underline{T = 1}$$

$$\tau_D = 1 + 1.5 + 2 + 1.5 = \underline{\underline{6 \text{ nsec}}}$$

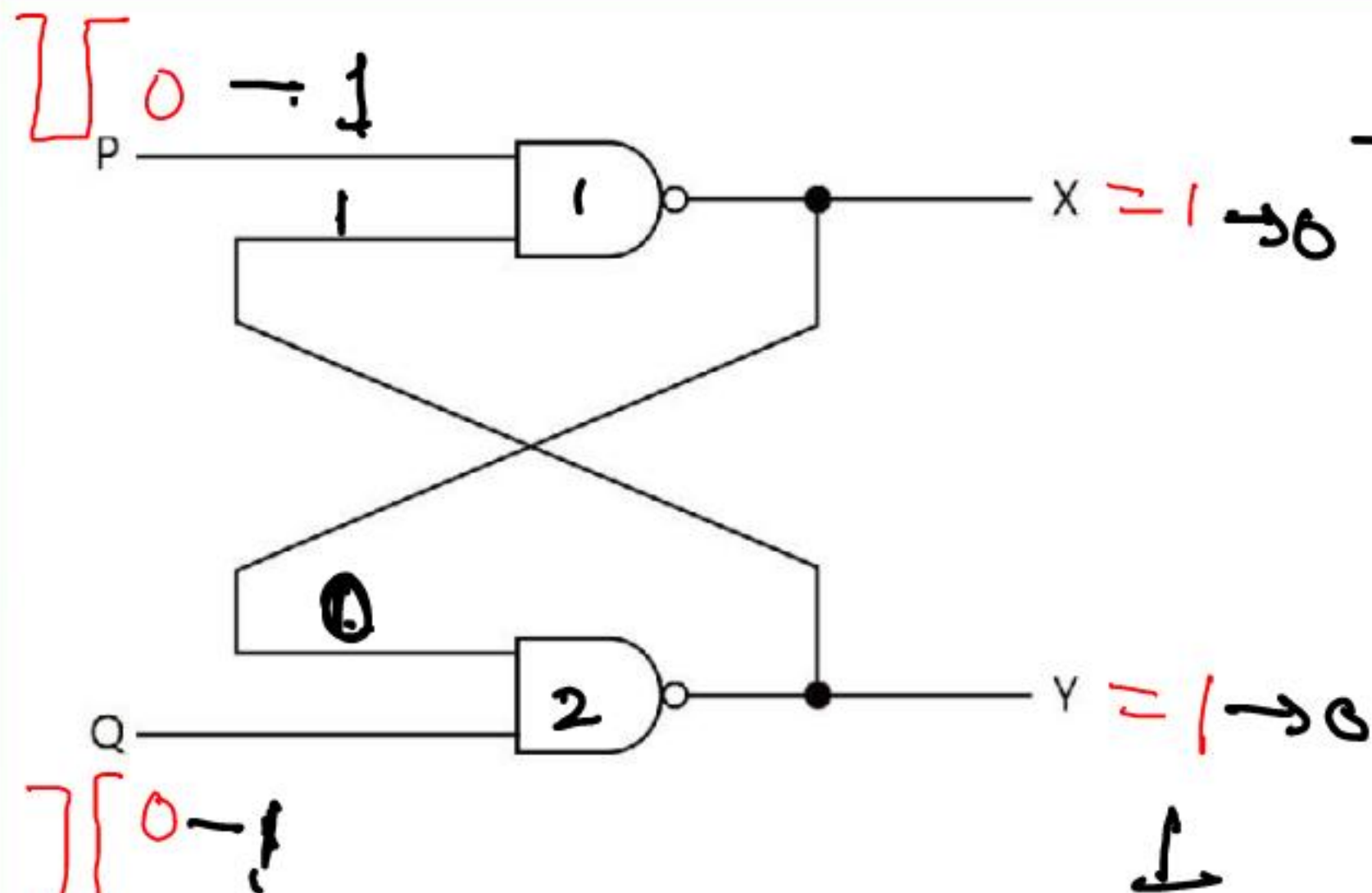
$$\tau_D = \underline{\underline{6 \text{ nsec}}}$$



Q. In the latch circuit shown, the NAND gates have non-zero but unequal propagation delays. The present input condition is $P = Q = '0'$. If the input condition is changed simultaneously to $P = Q = '1'$, the outputs X and Y are

- A. $X = '1'$, $Y = 1$.
- ☒ B. either $X = '1'$, $Y = '0'$ or $X = '0'$, $Y = '1'$.
- C. either $X = '1'$, $Y = '1'$ or $X = '0'$, $Y = '0'$.
- D. $X = '0'$, $Y = '0'$.

[GATE-2017, 1 MARK]

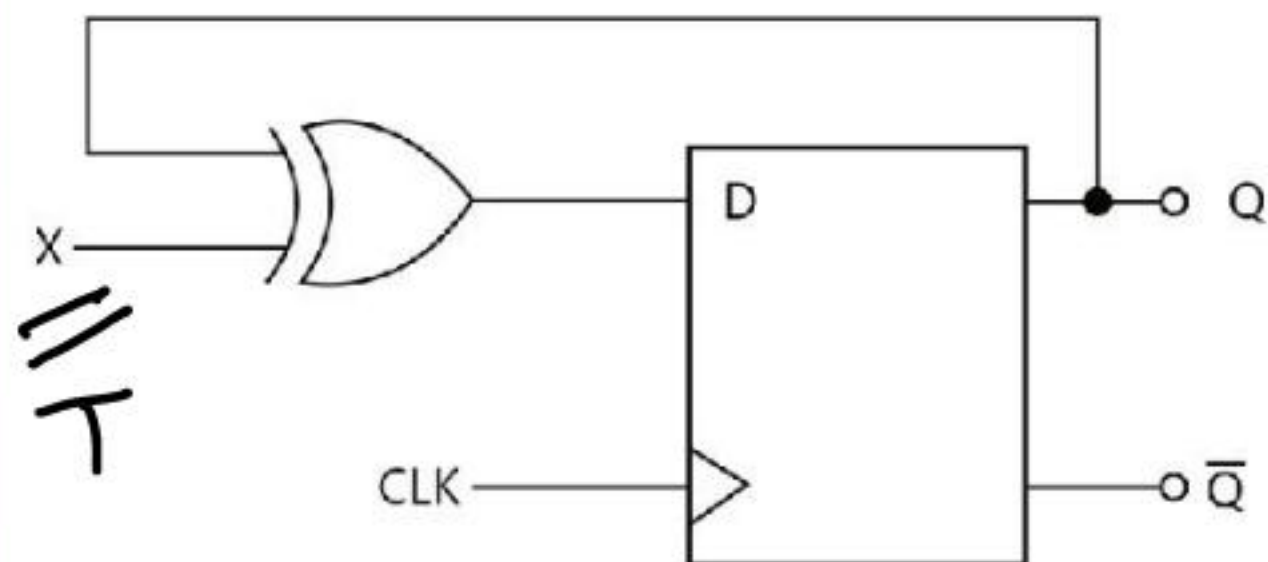


P	Q	X	Y
0	0	1	1
1	1	1	0
0	1	0	1

0	0	1
0	1	1
1	0	1
1	1	0

metastable

- Handwritten red annotations: $P=0$, $Q=1$.
- ① $\tau_2 < \tau_1$
 - ② $\tau_2 > \tau_1$



Q. The digital circuit shown in the figure works as

A. JK flip-flop

B. Clocked RS flip-flop

☒ C. T flip-flop

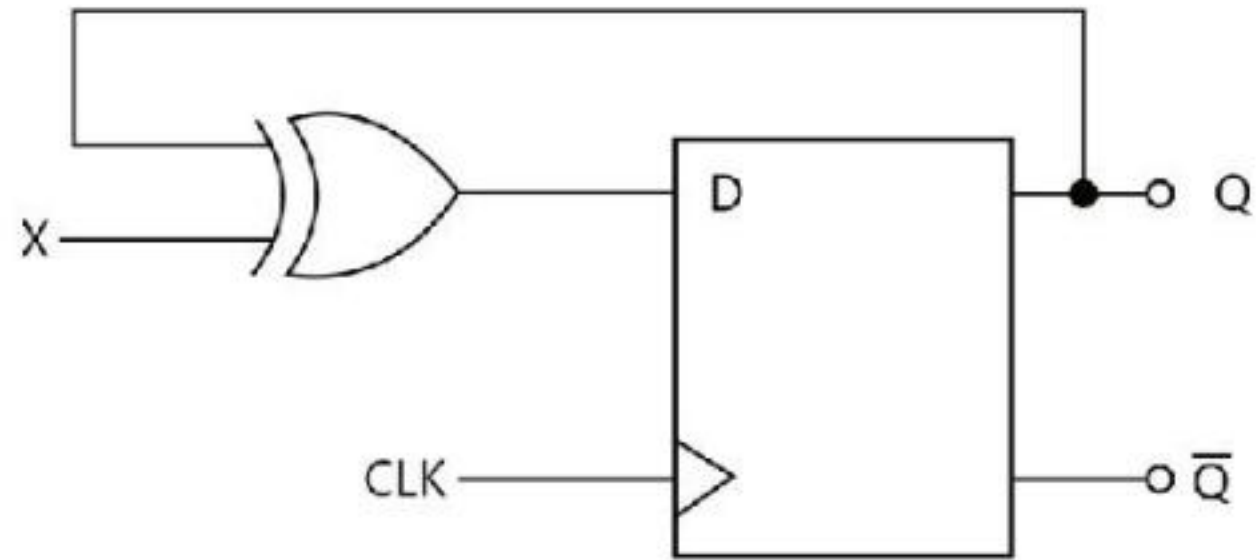
D. Ring counter

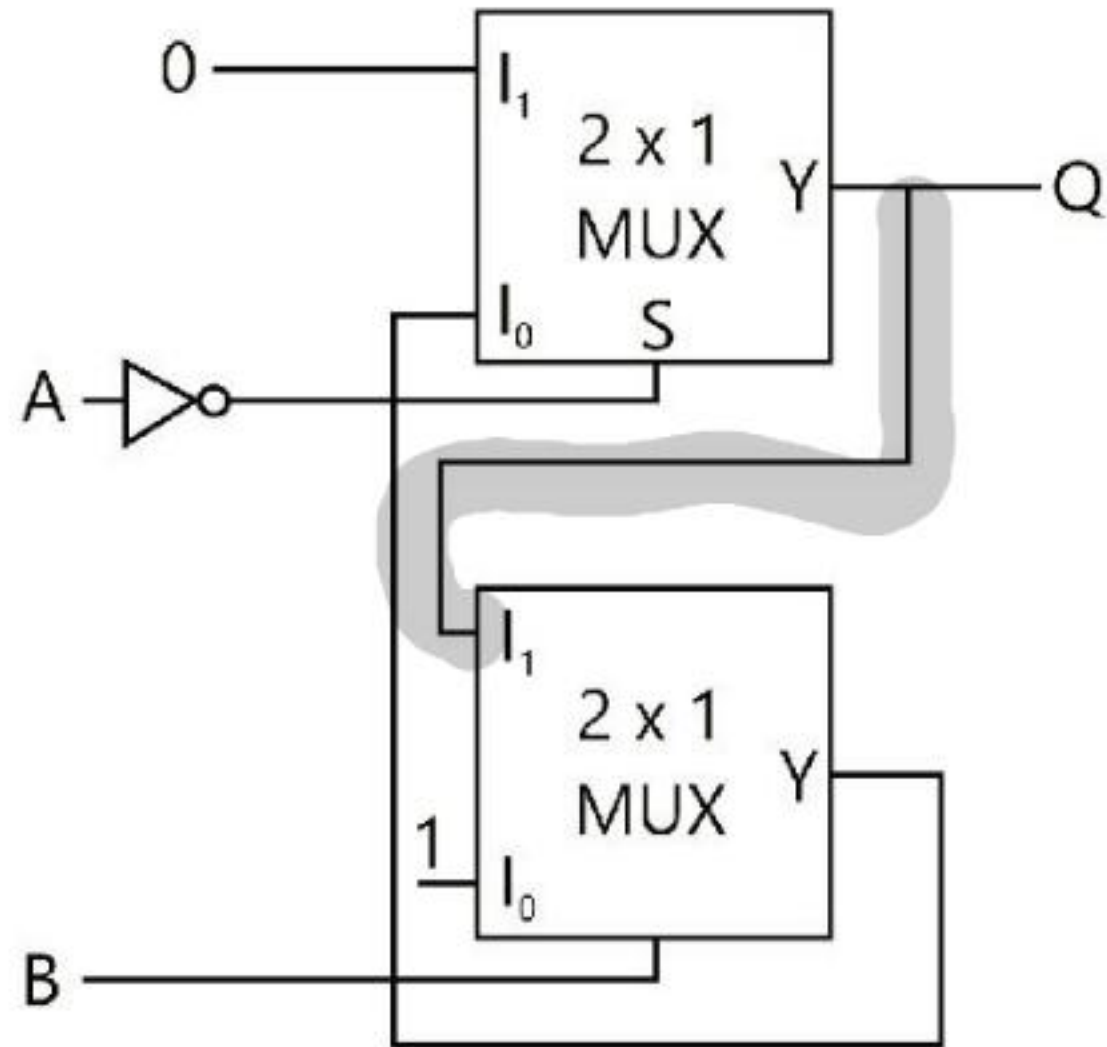
[GATE-2005, 1 MARKS]

$$D = X \oplus Q$$

$$Q(t+1) = D = X \oplus Q$$

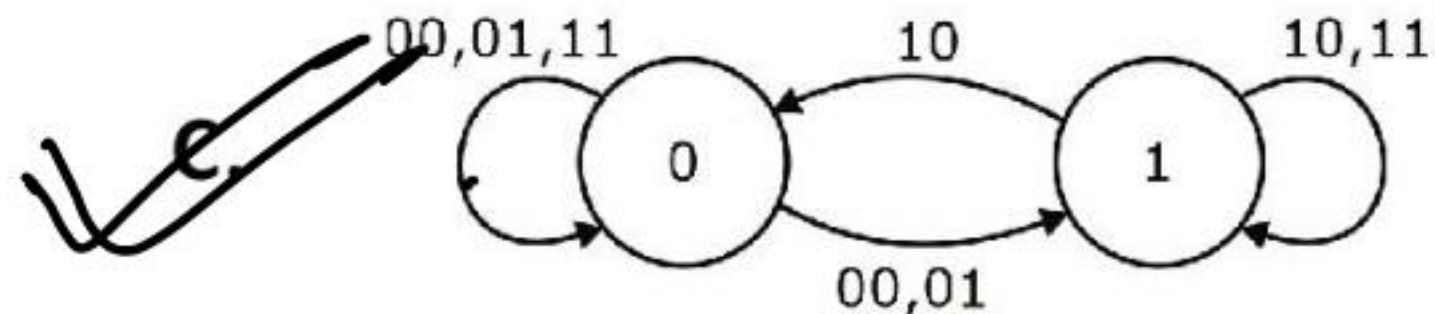
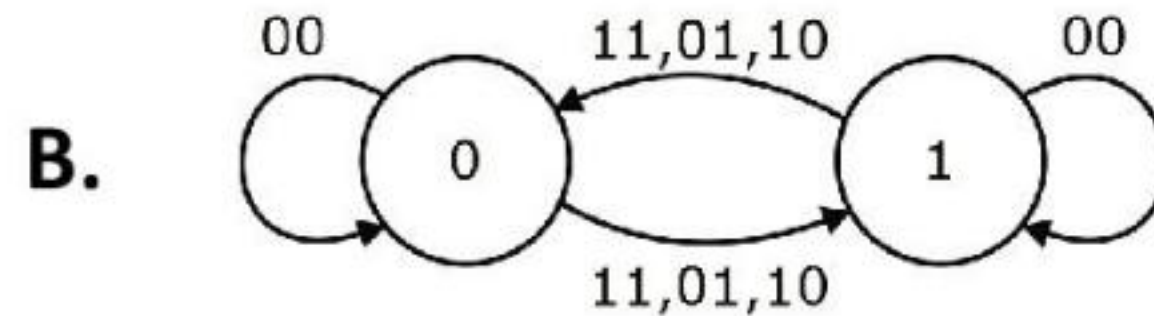
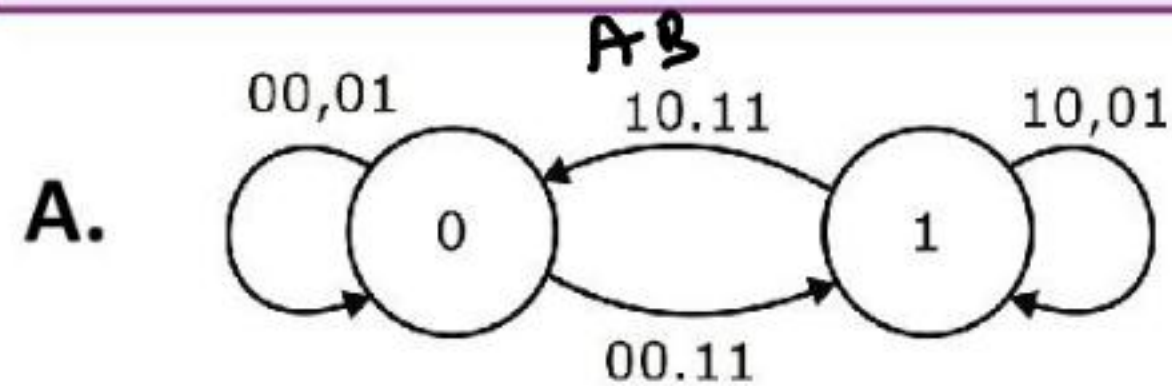
$$\left. \begin{aligned} Q(t+1) &= J\bar{Q} + \bar{K}Q \\ Q(t+1) &= T \oplus Q \\ Q(t+1) &= D \end{aligned} \right\}$$

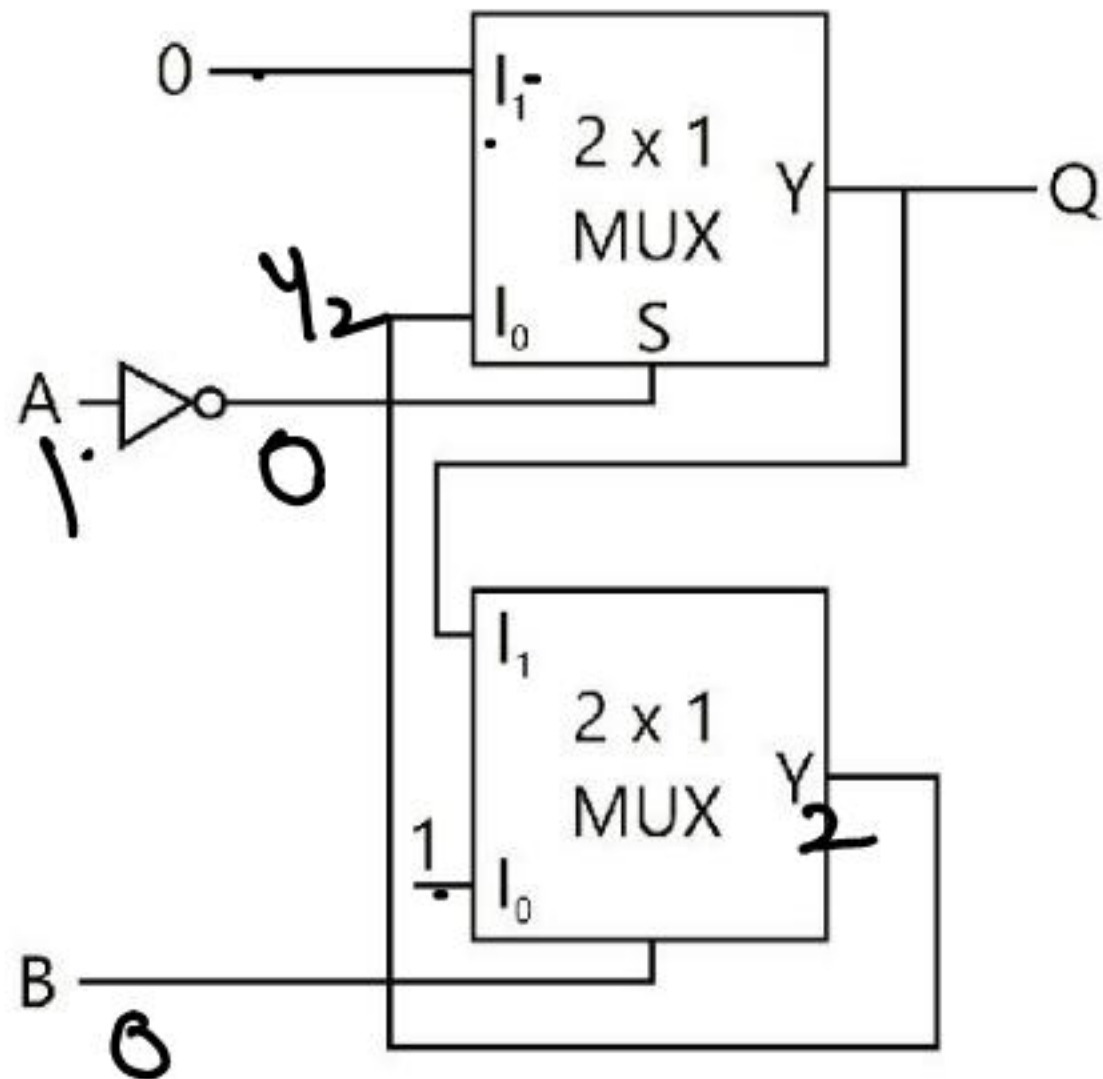




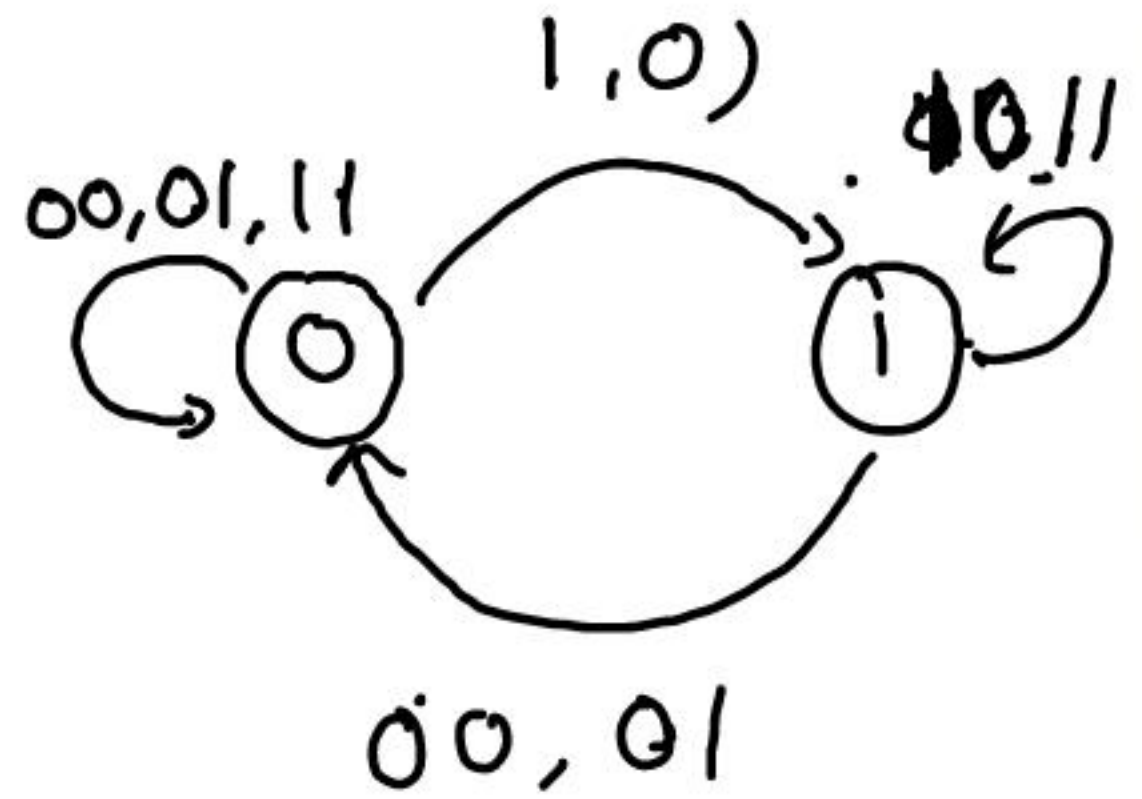
Q. Consider the following circuit:
Sketch the state diagrams

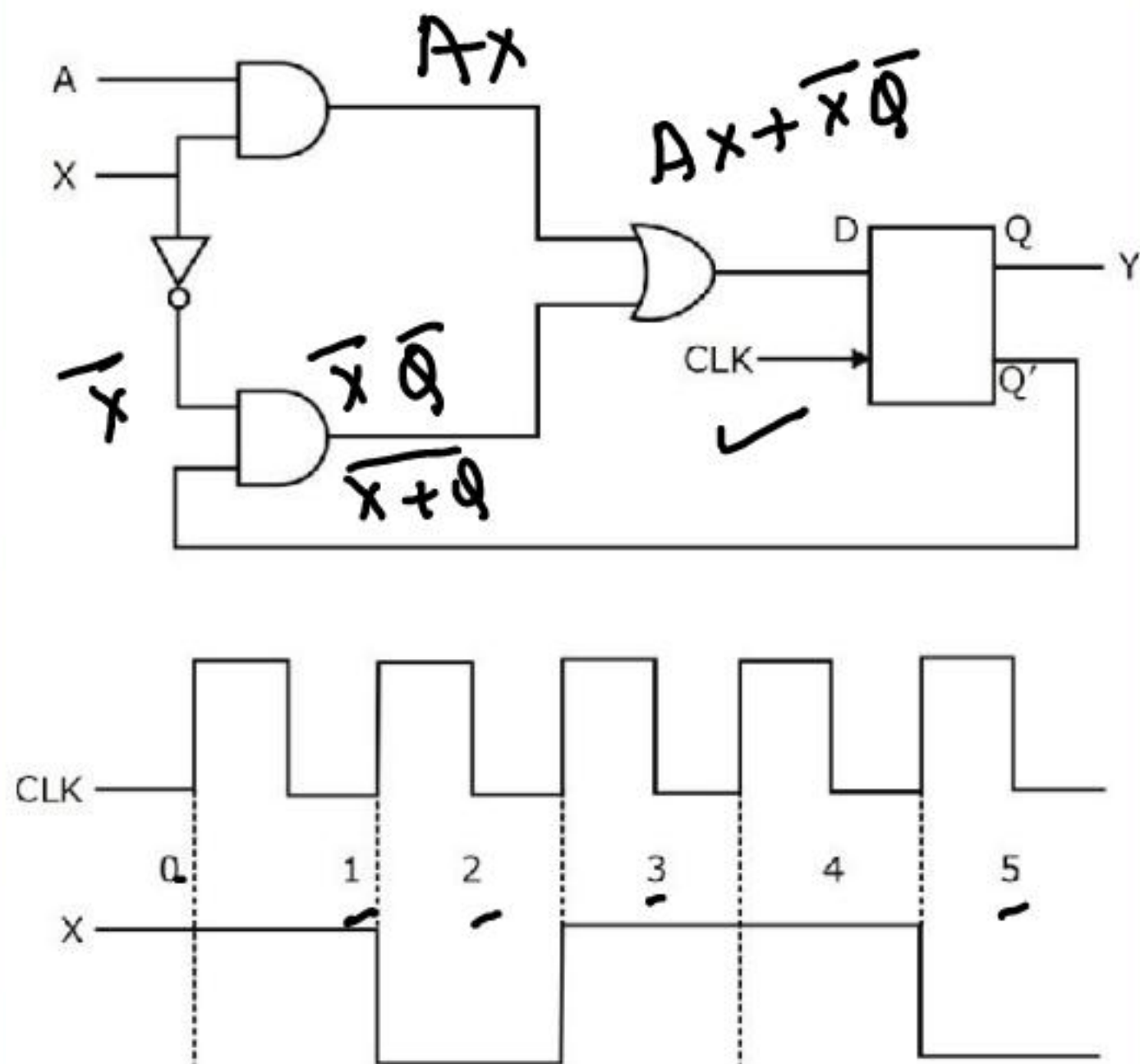
[GATE-2014, 1 MARKS]





A	B	Q	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1



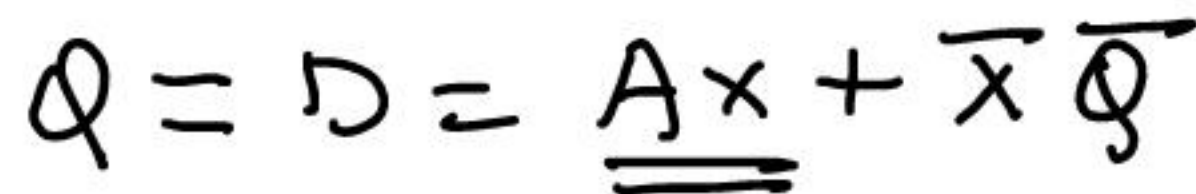


Q. Consider the following circuit involving a positive edge triggered D FF.

Consider the following timing diagram. Let A_i represent the logic level on the line A in the i th clock period.

Let A' represent the complement of A. The correct output sequence on Y over the clock periods 1 through 5 is

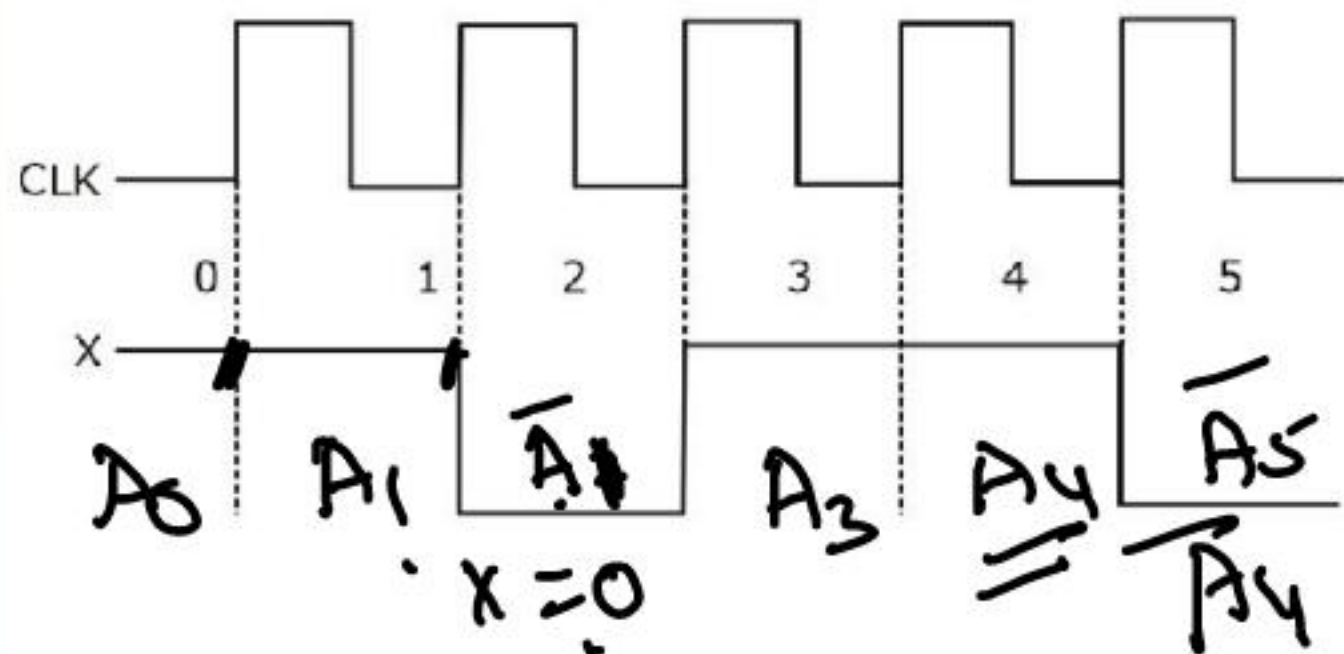
- ☒ A. $A_0 A_1 A_1' A_3 A_4$
- B. $A_0 A_1 A_2' A_3 A_4$
- C. $A_1 A_2 A_2' A_3 A_4$
- D. $A_1 A_2' A_3 A_4 A_5'$

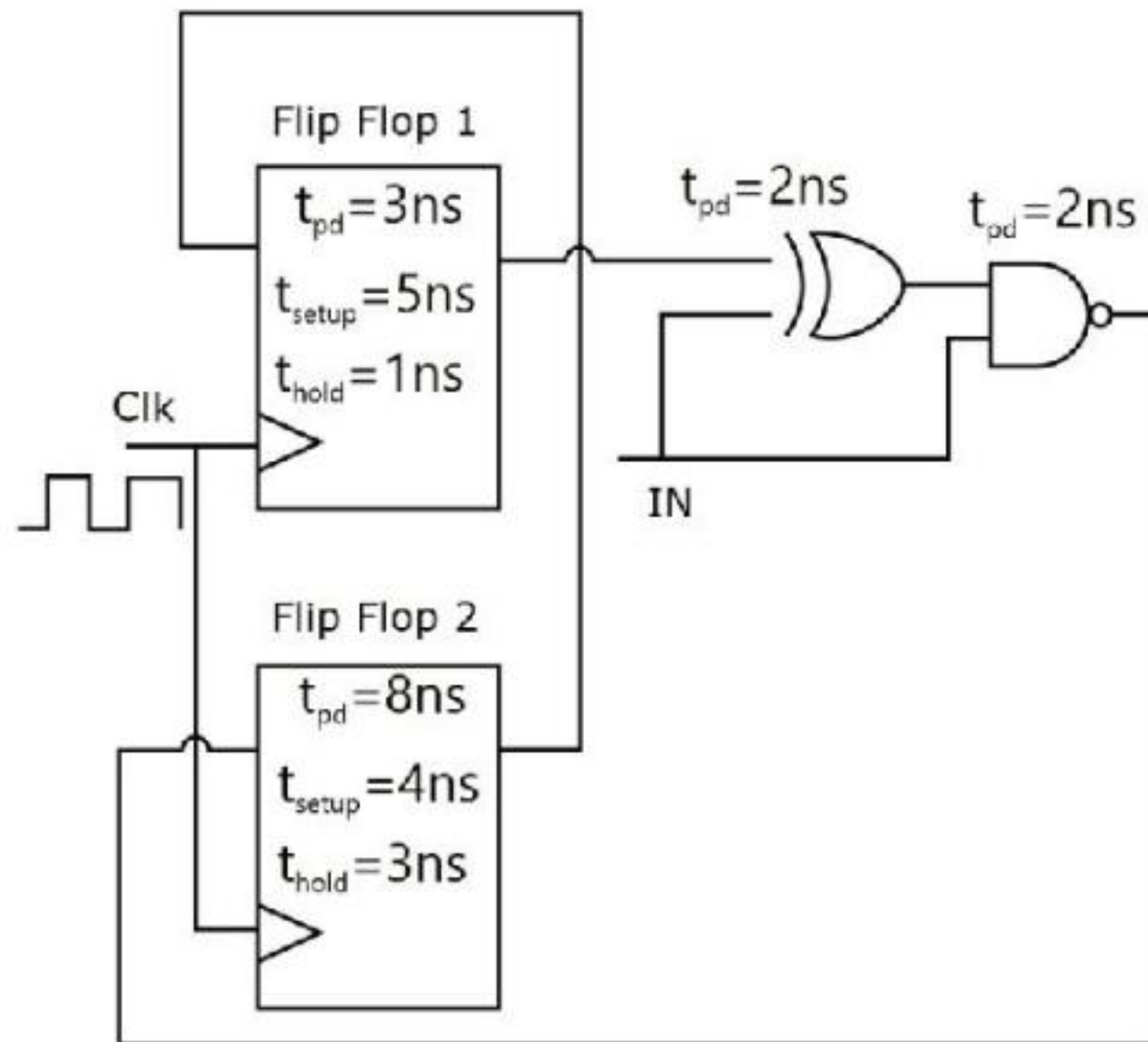


$$Q = A$$

$$Q = \overline{Q} = \overline{A}$$

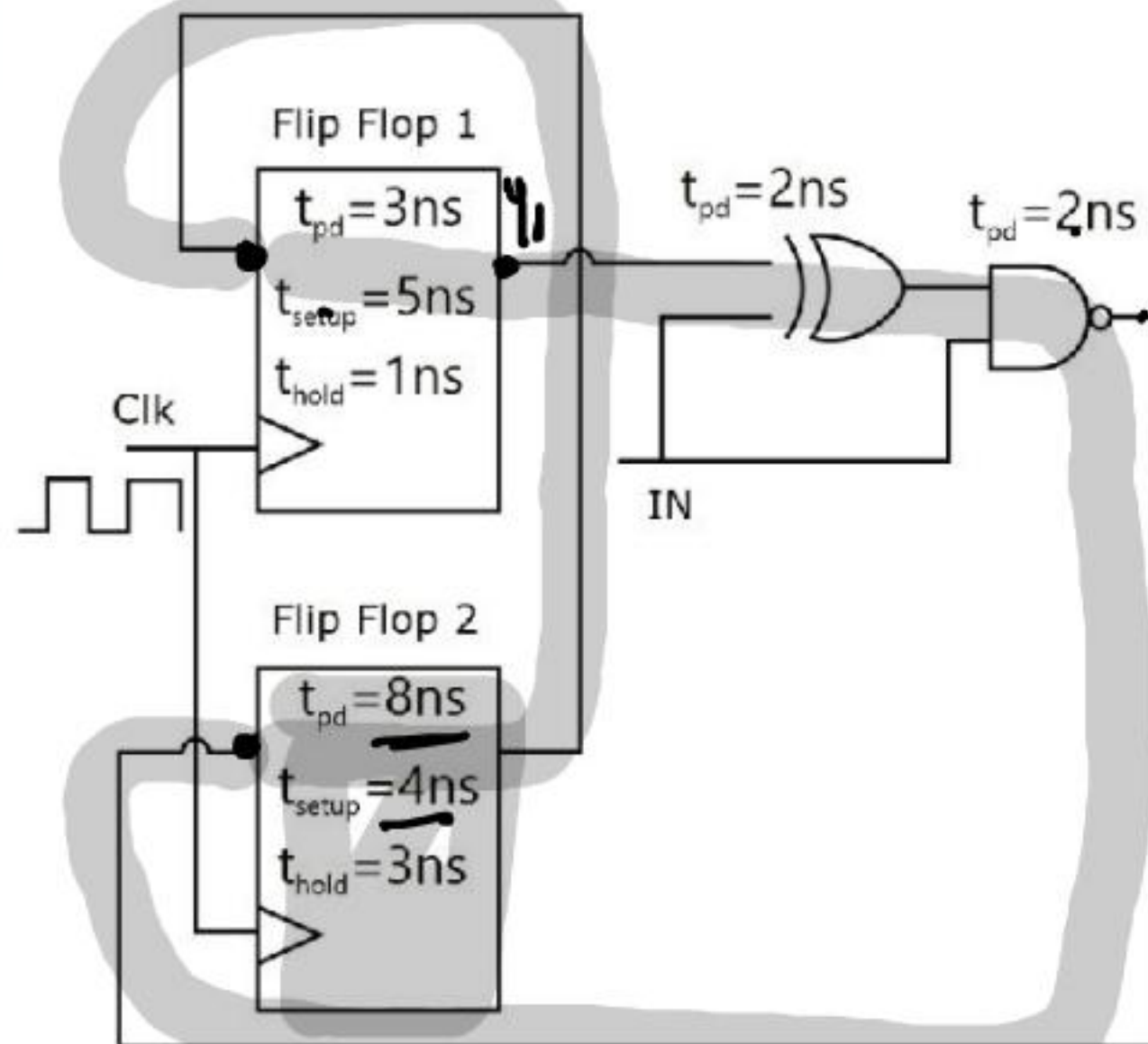
$$A_0, A_1, \overline{A}, A_3, A_4, \overline{A_5}$$





Q. For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is _____ MHz

[GATE-2020, 2 MARKS]



$$\tau_{d_2} = \tau_{pd_1} + \tau_{com_1} + \tau_{s_2} =$$

$$3 + 4 + 4 = \underline{\underline{11 \text{ nsec}}}$$

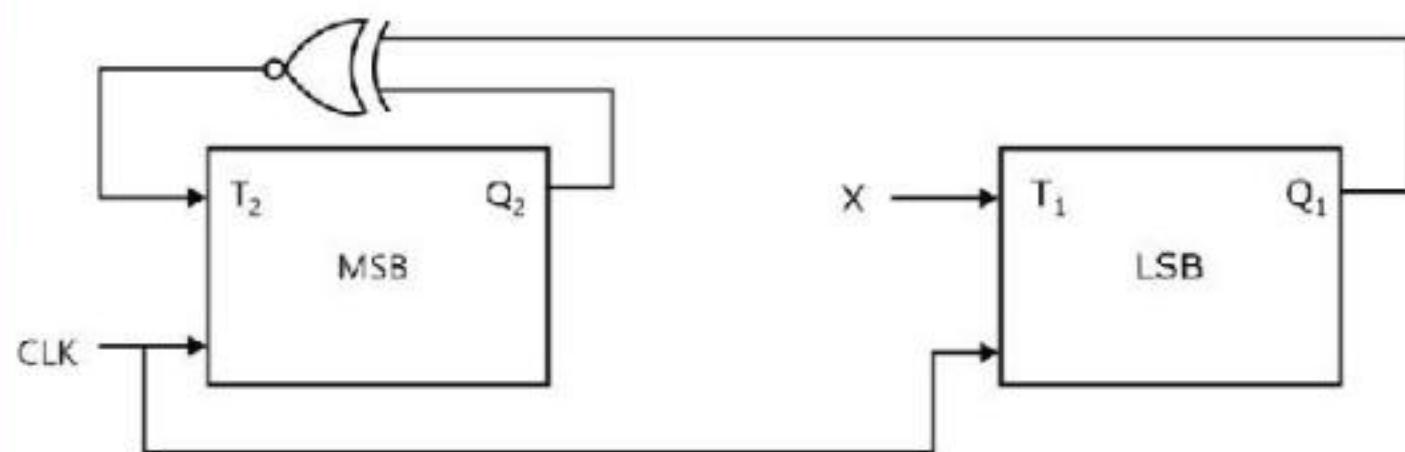
$$\tau_{d_1} = \tau_{pd_2} + \tau_{com_2} + \tau_{s_1}$$

$$= 8 + 0 + 5 = \underline{\underline{13 \text{ nsec}}}$$

$$\tau_d = \underline{\underline{13 \text{ nsec}}}$$

$$f_{\max} = \frac{1}{13 \times 10^{-9}} = \frac{1000 \times 10^6}{13}$$

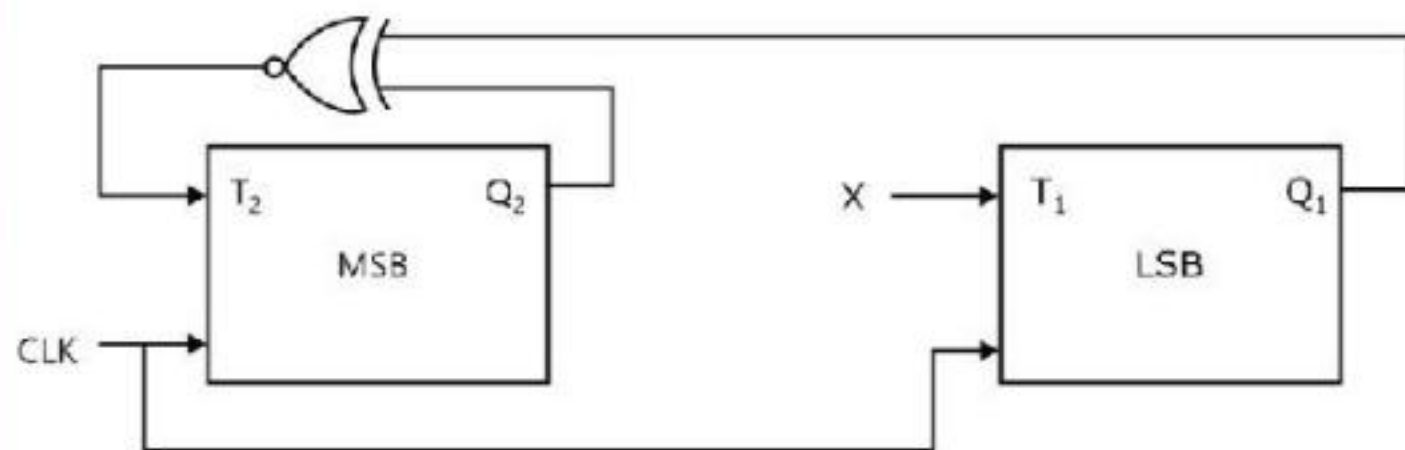
$$= \frac{1000}{13} \text{ MHz}$$



Q. Consider the partial implementation of a 2-bit counters using T flip flops following the sequence 0-2-3-1-0, as shown below

To complete the circuit, the input X should be

- A. Q'_2
- B. $Q_2 + Q_1$
- C. $(Q_1 \oplus Q_2)'$
- ☒ D. $(Q_1 \oplus Q_2)$

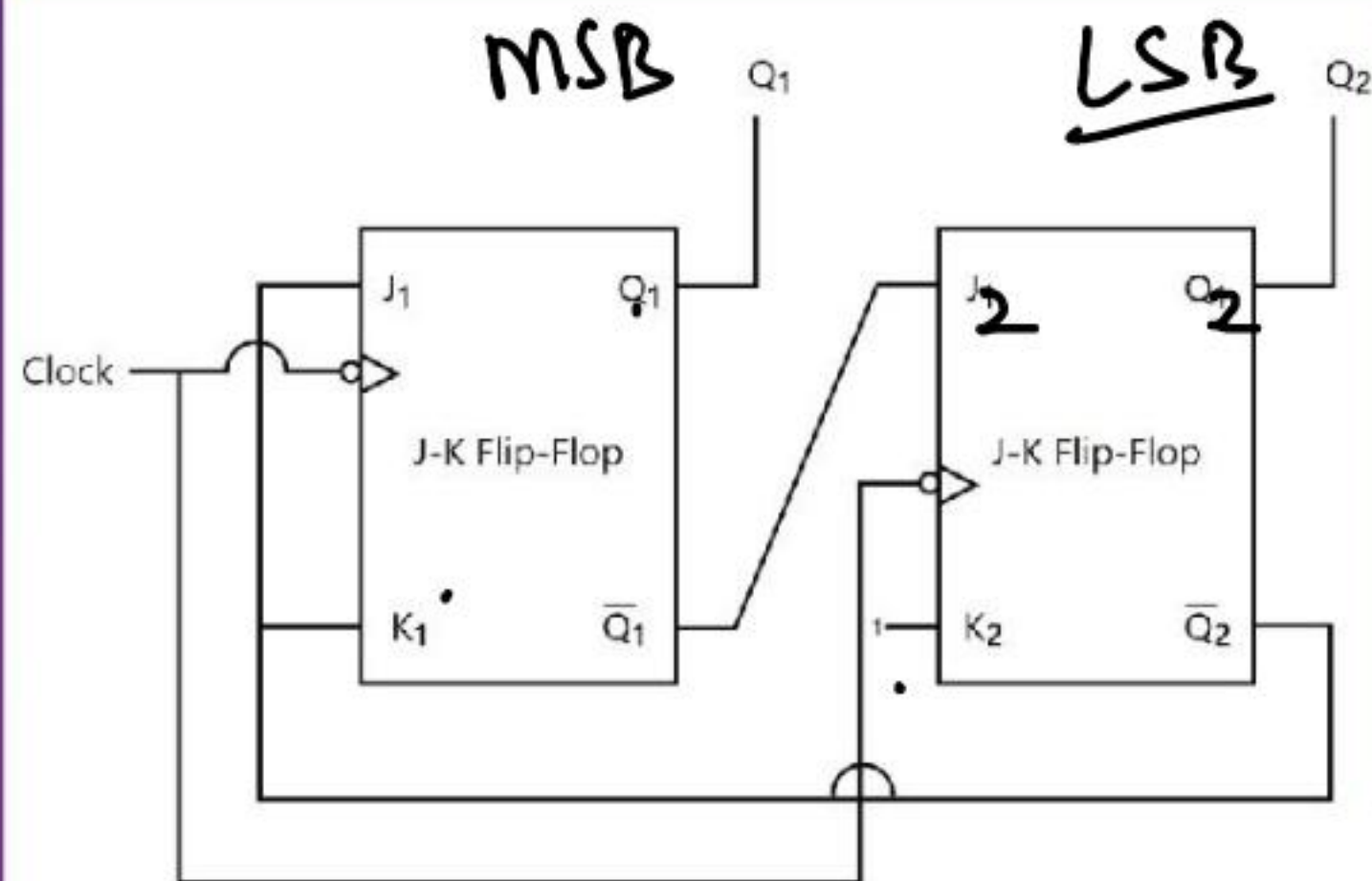


Q_2	Q_1	Q_2	Q_1	T_2	T_1
0	0	1	0	1	0
1	0	1	1	0	1
1	1	0	1	1	0
0	1	0	0	0	1
0	0	1	0	1	0

$$T_1 = Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1$$

$$= \underline{Q_2 \oplus Q_1}$$

Q	$Q (+1)$	T
0	1	1
1	0	1
0	0	0
1	1	0



Q. What are the counting stages (Q_1, Q_2) for the counter shown in the figure below?

- ☒ A. 11, 10, 00, 11, 10,
- B. 01, 10, 11, 00, 01,
- C. 00, 11, 01, 10, 00,
- D. 01, 10, 00, 01, 10,

10 min

$$J_1 = K_1 = \overline{Q_2}$$

$$J_2 = \overline{Q_1}, \quad K_2 = 1$$

$Q_2 \ Q_1$	J_2 $\overline{Q_1}$	K_2 1	J_1 $\overline{Q_2}$	K_1 $\overline{Q_2}$	$Q_2 \ Q_1$
1 0	1	1	1	1	1 1
1 1	0	1	0	0	0 1
0 1	0	1	1	1	0 0
0 0					

$Q_1 \ Q_2$

00 - 11 - 10 - 00

0-1-3-6-4

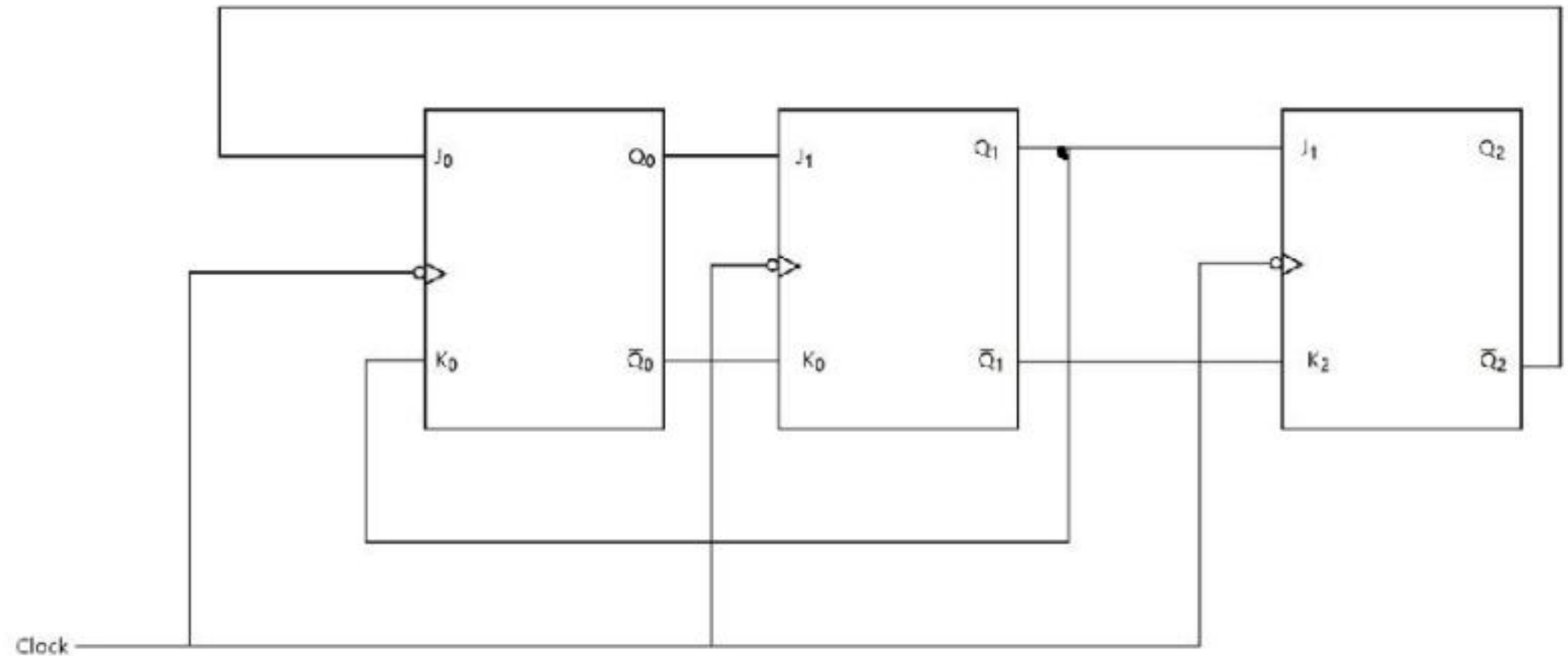
Q. Modulus of the following counter is

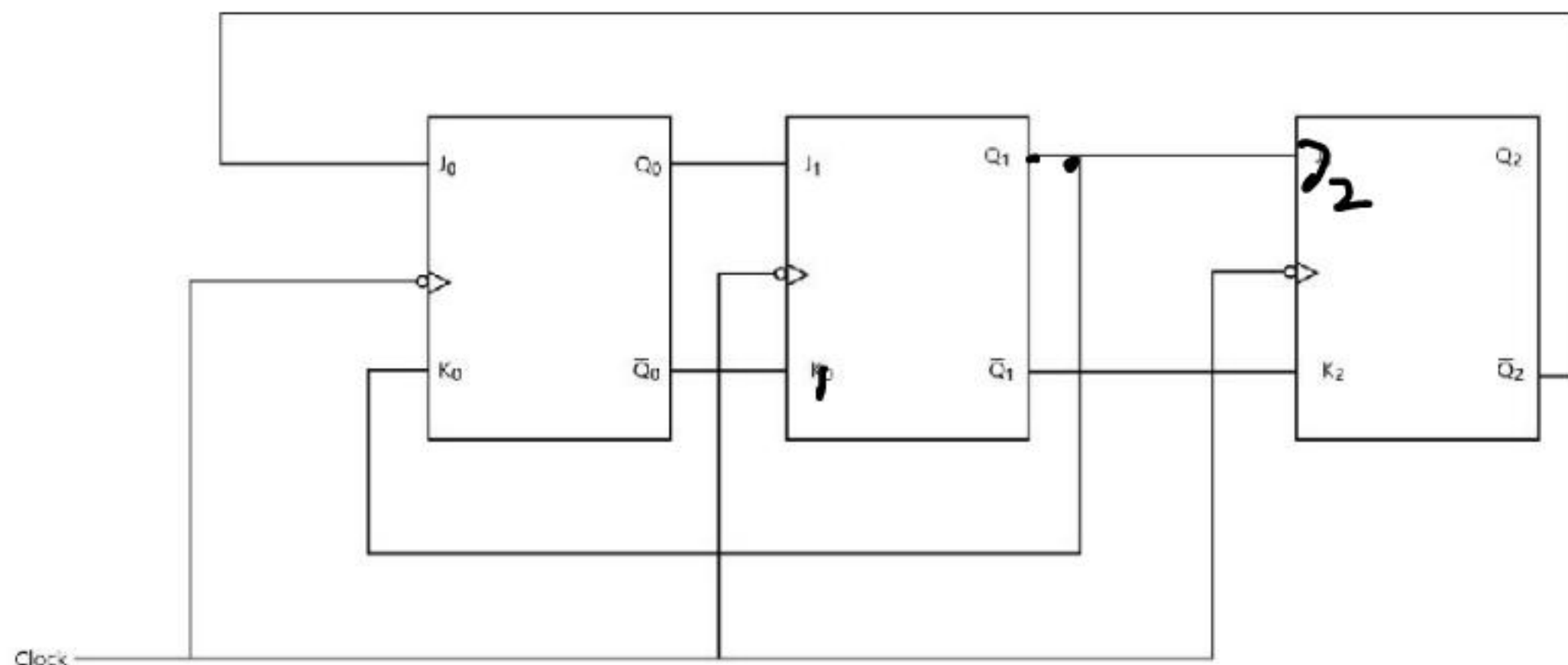
A. 3

✓ B. 5

C. 6

D. 8





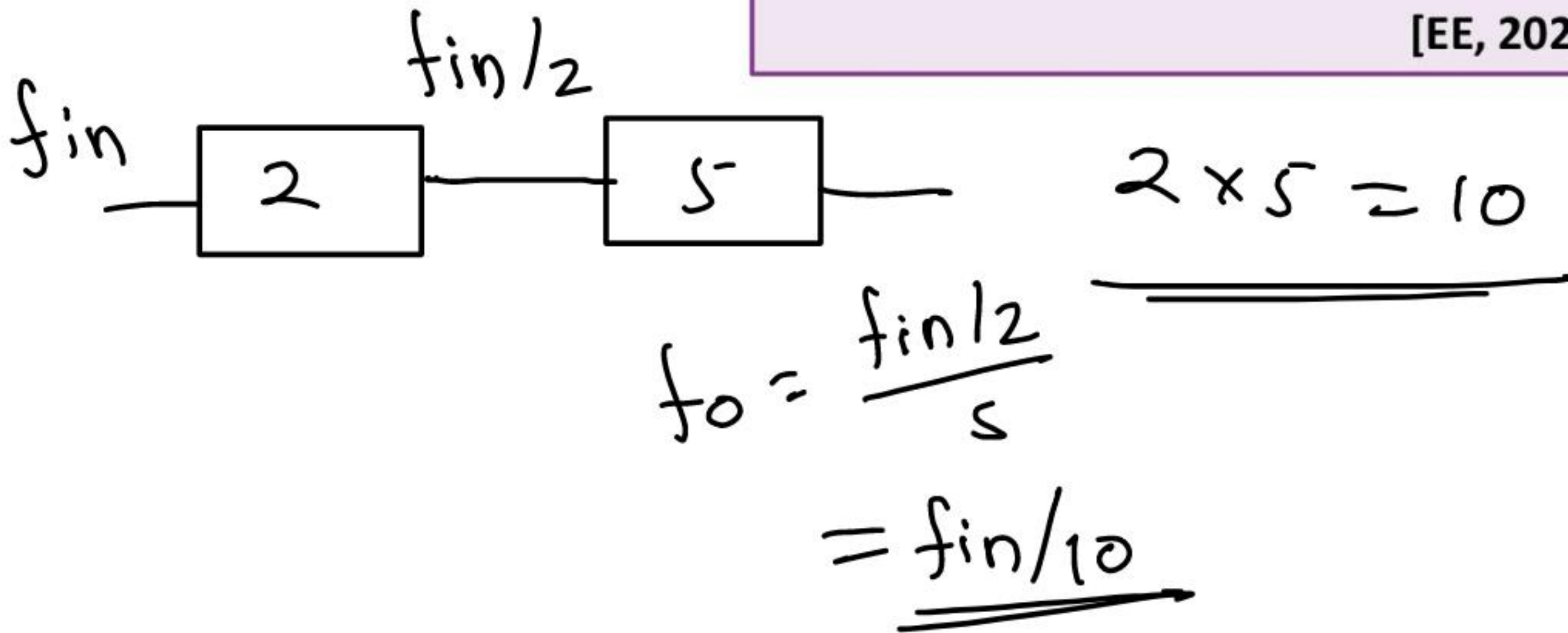
$Q_2 Q_1 Q_0$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$
1 0 0	0 1	0 1	1 0
0 0 1	0 1	1 0	1 0
0 1 1	1 0	1 0	1 1
1 1 0	1 0	0 1	0 1
1 0 0	0 1	0 1	0 0
0 0 0	1 0	0 1	0 0

mod-5

$$J_0 = \overline{Q_2}, K_0 = Q_1, J_1 = Q_0, K_1 = \overline{Q_0}, J_2 = Q_1, K_2 = \overline{Q_1}$$

Q. A MOD 2 and a MOD 5 up-counter when cascaded together results in a MOD _____ counter. (in integer)

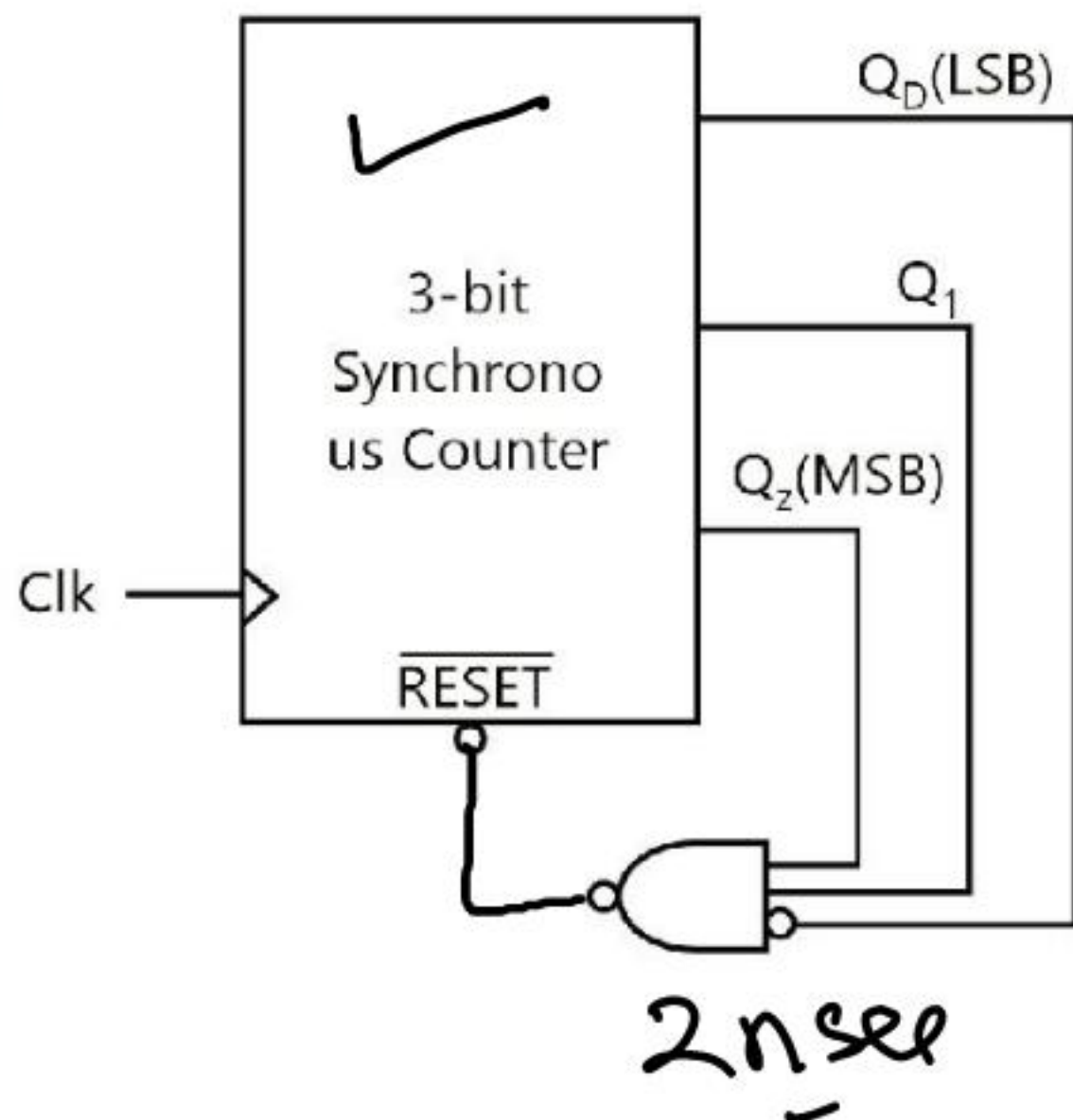
[EE, 2022 : 1 Mark]



0	000
0	100
1	001
1	101
2	010
2	110
3	011
3	111
0	
0	

~~Q.~~ The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0, 0, 1, 1, 2, 2, 3, 3, 0, 0,...) is 3.

0	00
1	01
2	10
3	11



Q. For the circuit shown in the figure, the delay of the bubbled NAND gate is 2 ns and that of the counter is assumed to be zero.

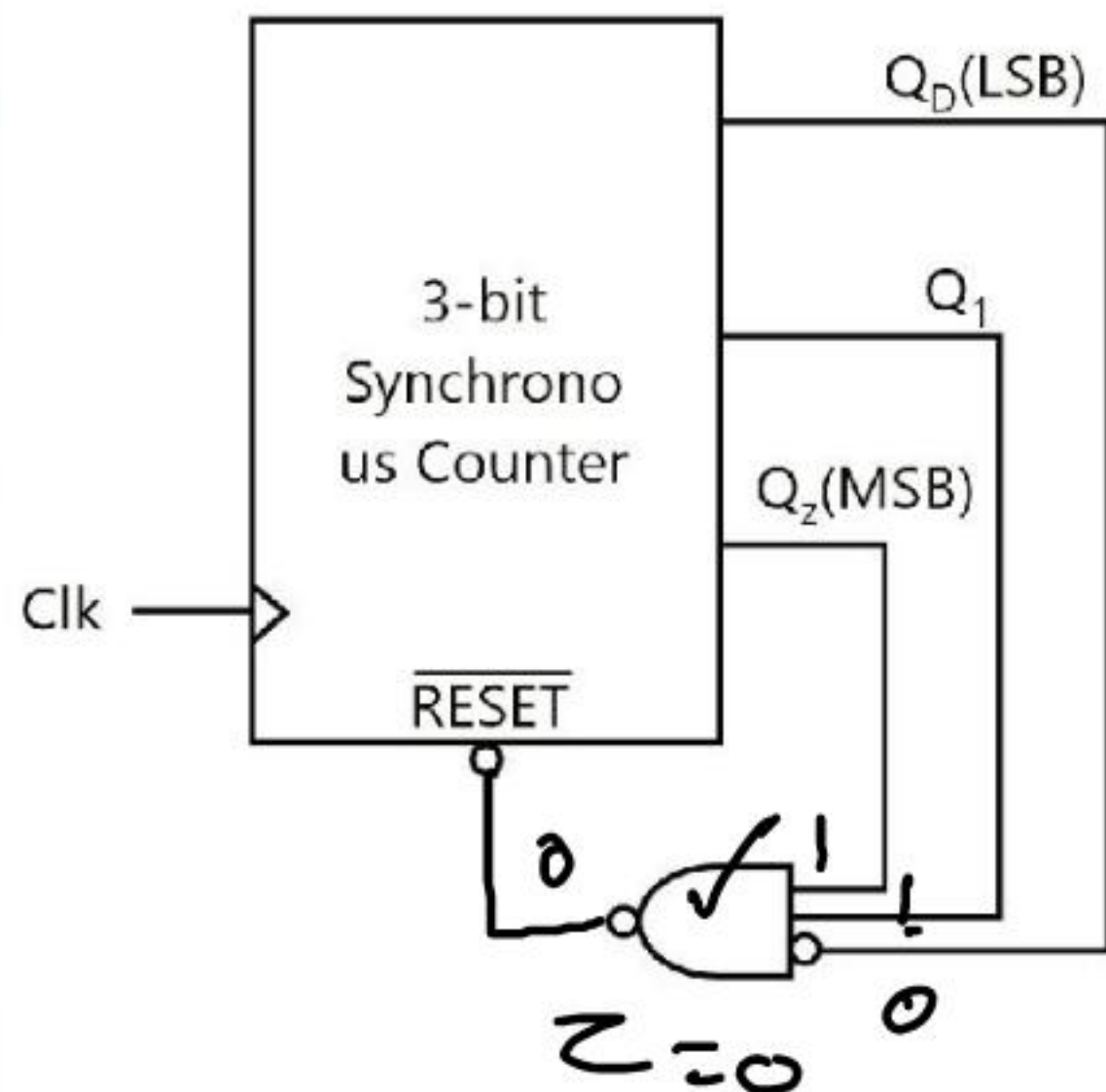
If the clock (Clk) frequency is 1 GHz, then the counter behaves as a

- A. mod-5 counter
- B. mod-6 counter
- C. mod-7 counter
- D. mod-8 counter

$$T = \frac{1}{f} = 10^{-9} \text{ sec}$$

$$T = \underline{\underline{1 \text{ nsec}}}$$

[GATE-2016, SET-3, 2-MARKS]



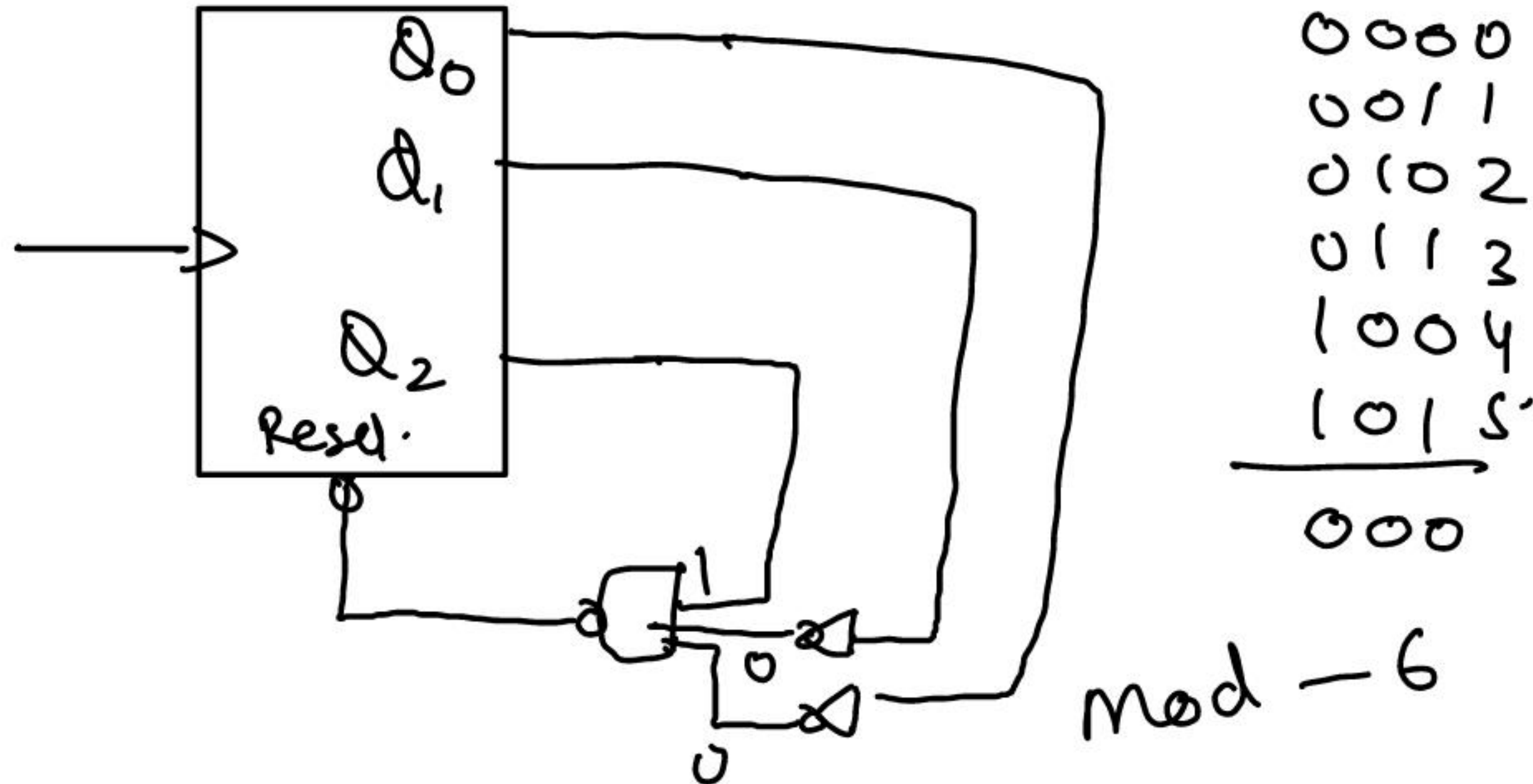
Asynchronous

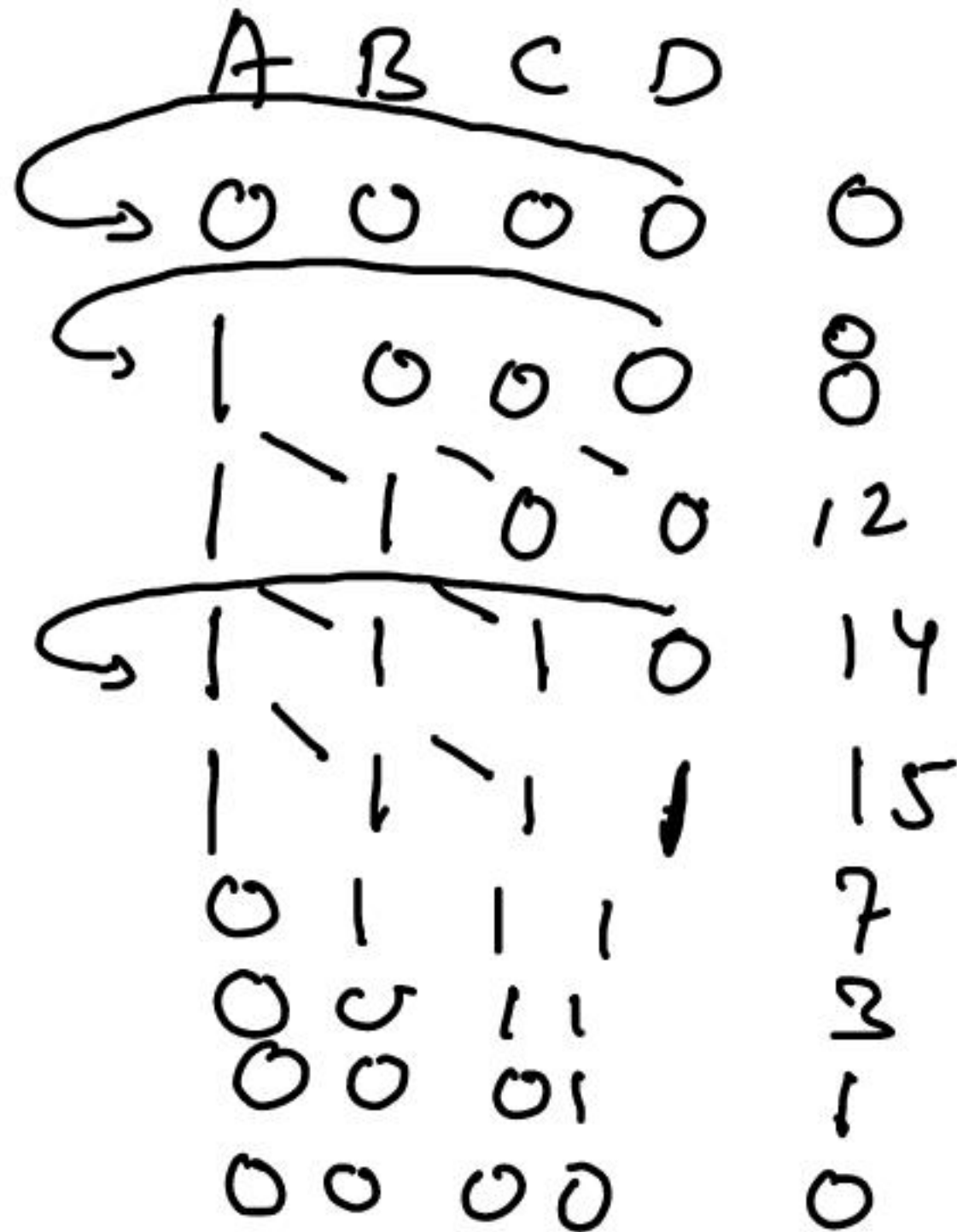
000	→ 0
001	→ 1
010	→ 2
011	→ 3
100	→ 4
101	→ 5
110	→ 6
111	→ 7
000	

$Q_2 Q_1 Q_0$
1 1 0

mod-6 Counter

Mod-8





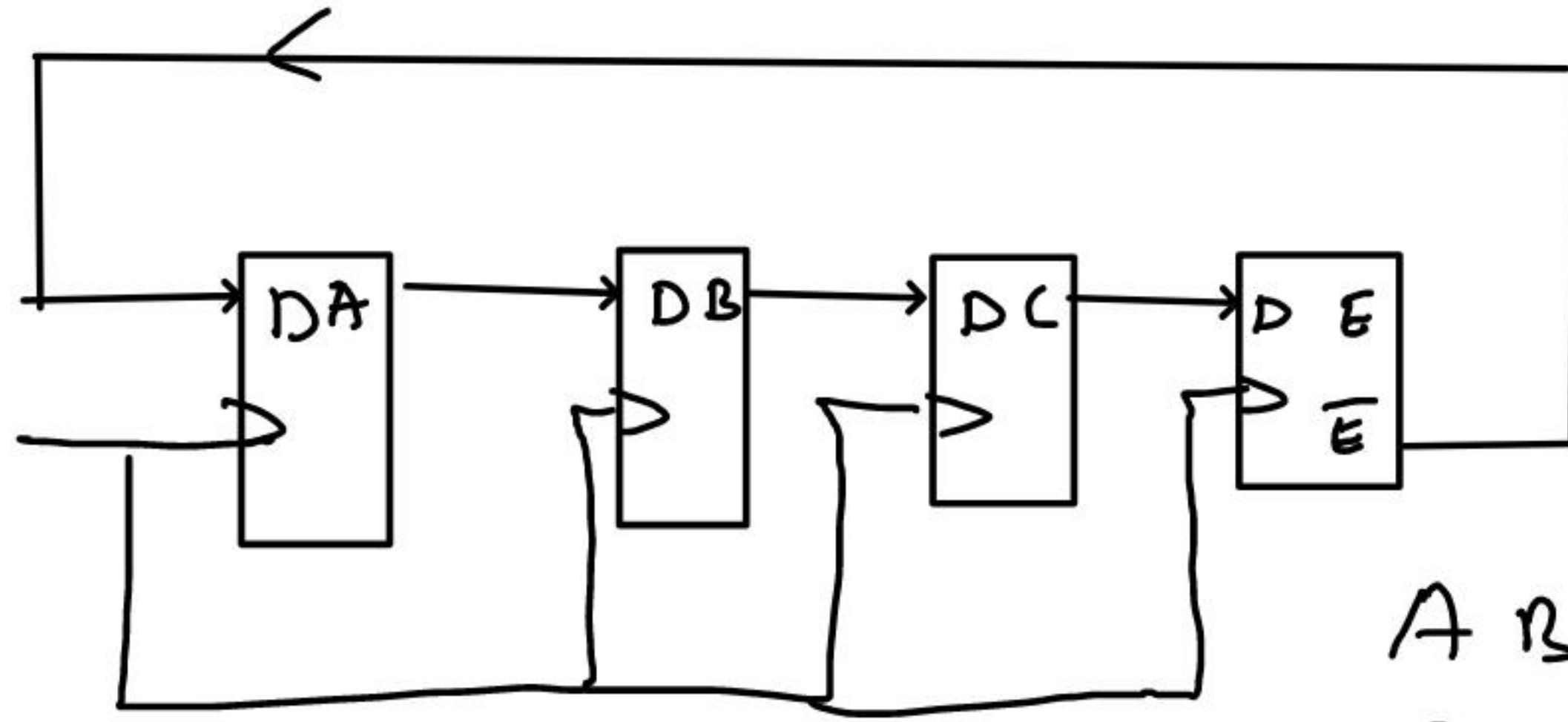
Q. Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is

A. 0, 1, 3, 7, 15, 14, 12, 8, 0

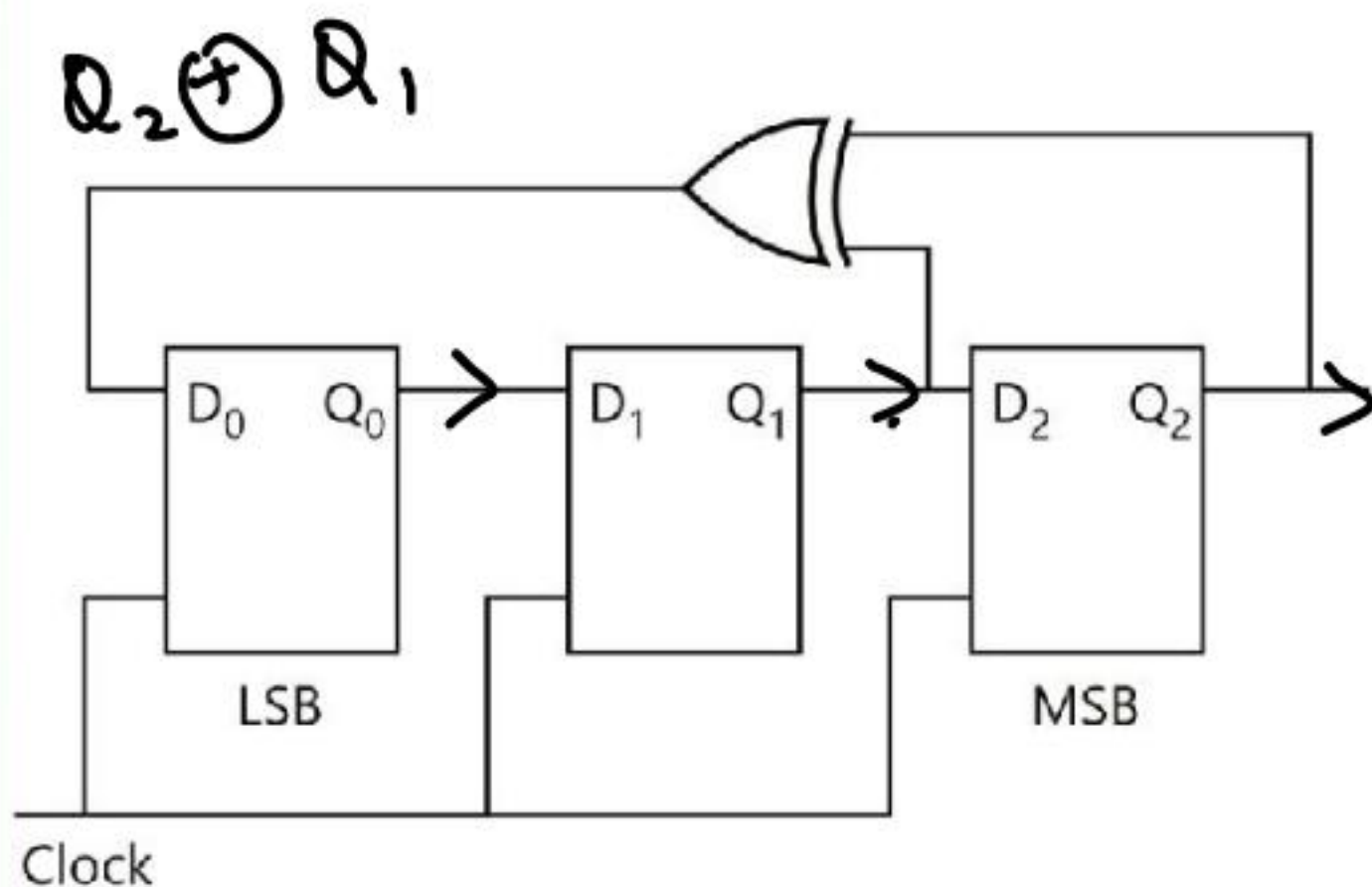
B. 0, 1, 3, 5, 7, 9, 11, 13, 15,

C. 0, 2, 4, 6, 8, 10, 12, 14, 0

☒ D. 0, 8, 12, 14, 15, 7, 3, 1, 0

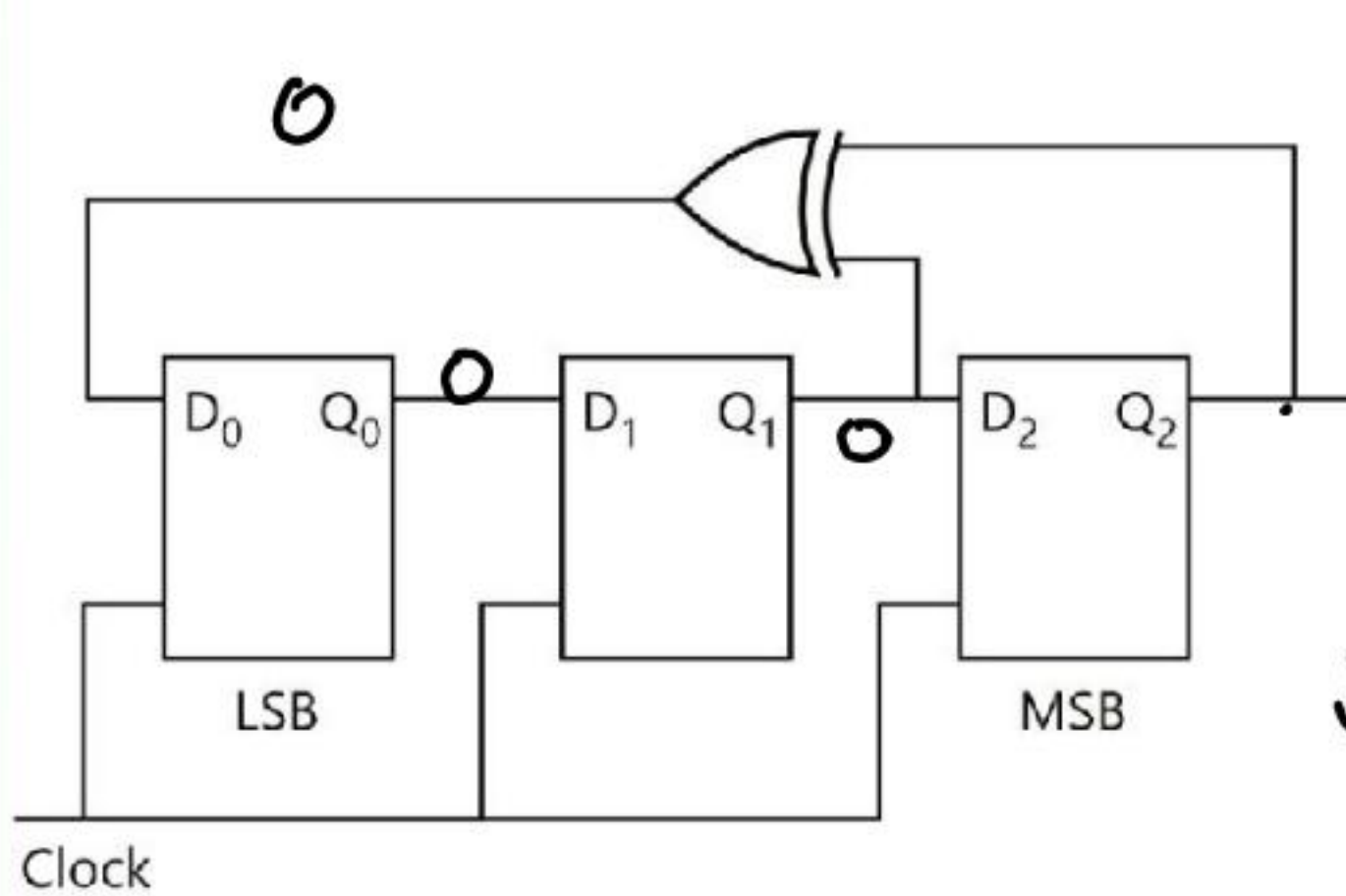


A	B	C	E
0	0	0	0
1	0	0	0
1	1	0	0

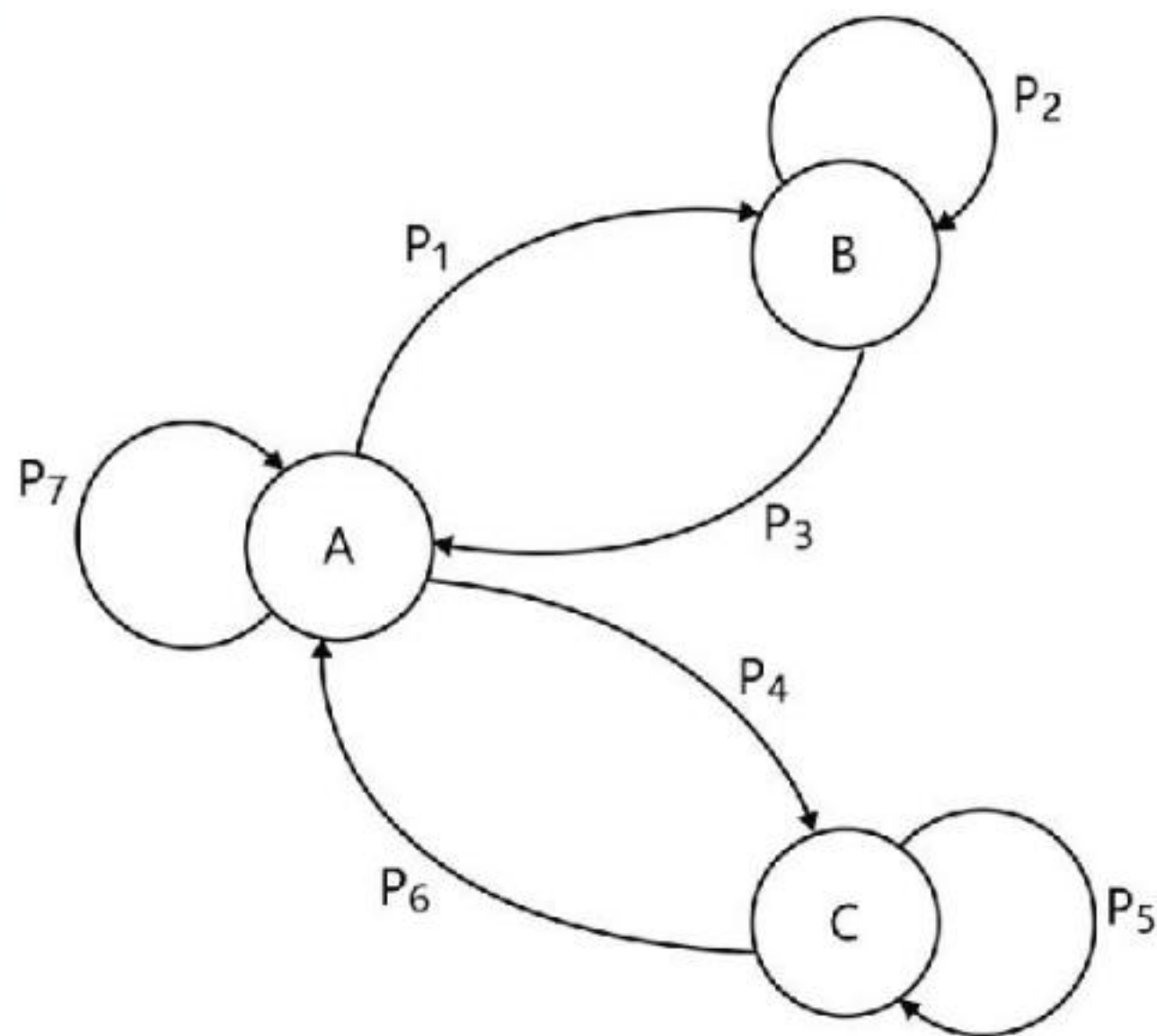


Q. Consider the circuit given below with initial state $Q_0 = 1, Q_1 = Q_2 = 0$. The state of the circuit is given by the value $\underline{4Q_2} + \underline{2Q_1} + \underline{Q_0}$. Which one of the following is the correct state sequence of the circuit?

- A. 1,3,4,6,7,5,2
- ✓ B. 1,2,5,3,7,6,4
- C. 1,2,7,3,5,6,4
- D. 1,6,5,7,2,3,4



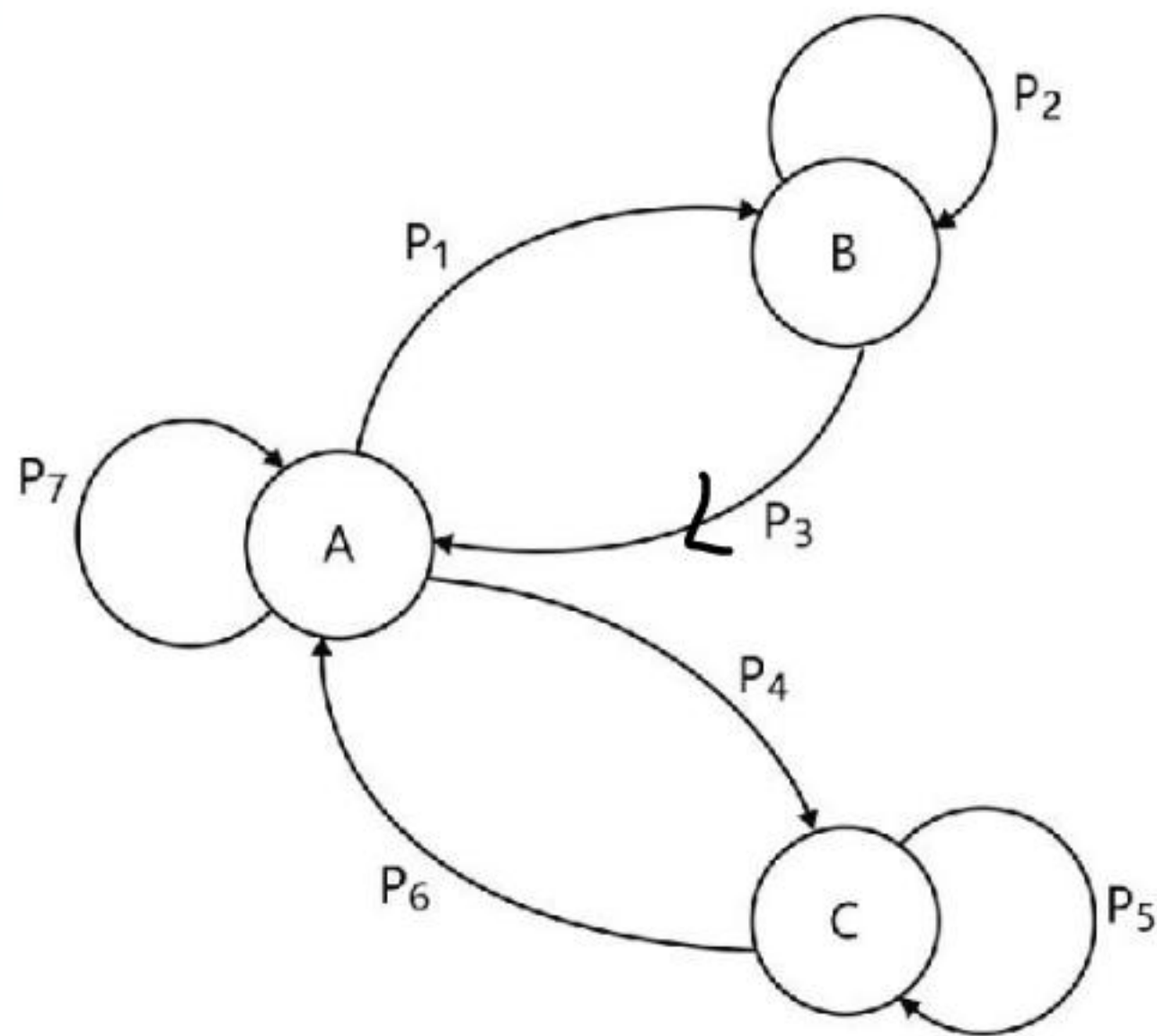
	Q_2	Q_1	Q_0	$D_0 = Q_2 \oplus Q_1$	Q_2	Q_1	Q_0
1	0	0	1	0	0	1	0
2	0	1	0	1	0	1	0
3	1	0	1	1	1	0	1
4	0	1	1	1	0	1	1
5	1	1	1	0	1	1	1
6	1	1	0	0	1	1	0
7	1	0	0	1	1	0	0
8	0	0	1	0	0	0	1



Q. A state transition diagram with states A, B and C, and transition probabilities $p_1, p_2, \dots, p_7$ is shown in the figure (e.g., p_1 denotes the probability of transition from state A to B). For this state diagram, select the statement (s) which is/are universally true.

- MSQ.
- ✓ A. $p_2 + p_3 = p_5 + p_6$
 - B. $p_1 + p_3 = p_4 + p_6$
 - C. $p_1 + p_4 + p_7 = 1$ ✓
 - D. $p_2 + p_5 + p_7 = 1$

[GATE 2022, 2 MARKS]



A $P_1 + P_4 + P_7 = 1$

B $P_3 + P_2 = 1$

C $P_5 + P_6 = 1$

$$P_2 + P_3 = P_5 + P_6$$

┌ ┌ ┌ ┌

4 digit BCD

$$R = \frac{V_{FS}}{10^4 - 1} = \frac{15}{9999}$$

=

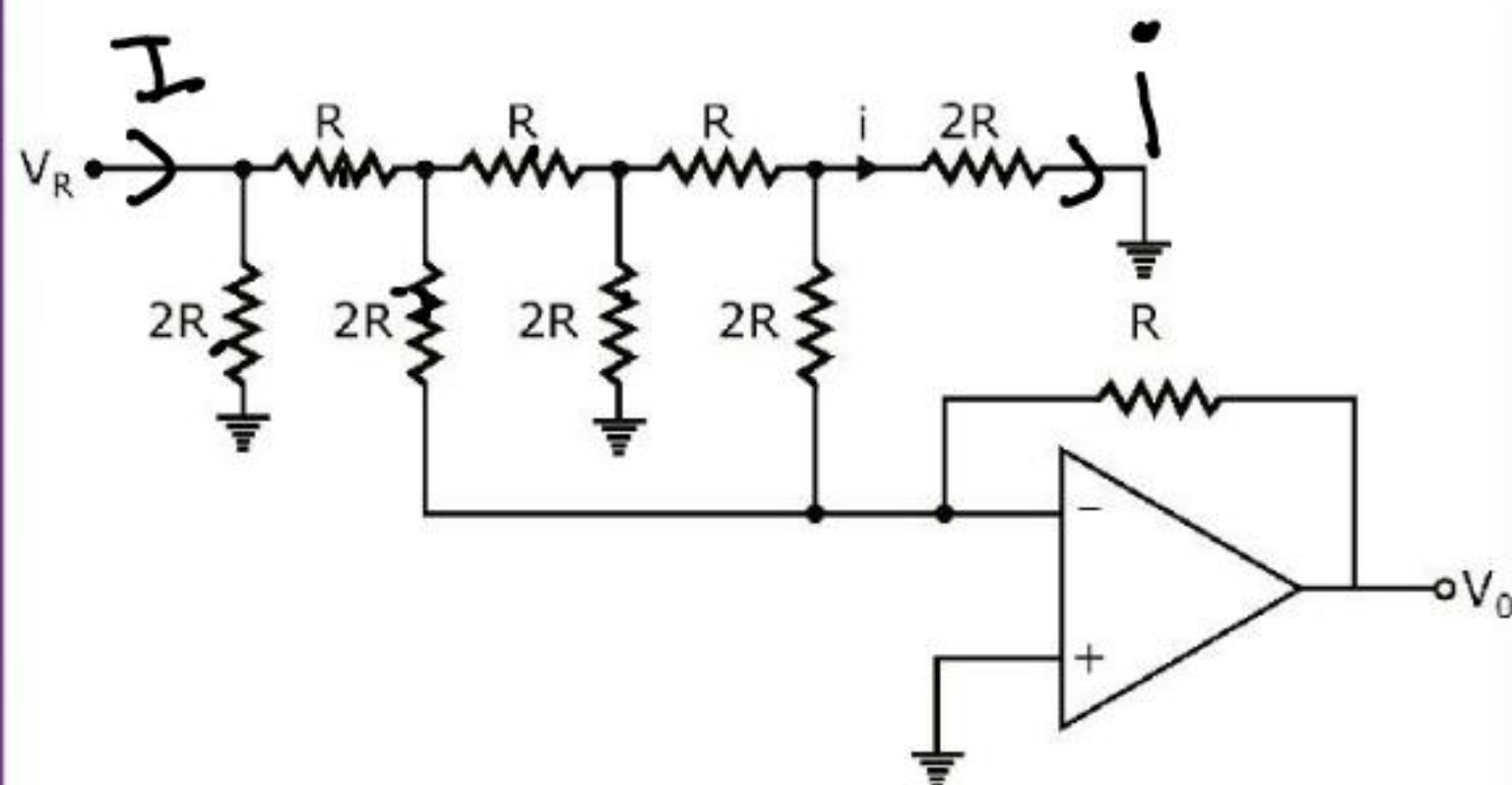
Q. Certain 16-bit BCD digital to analog converter has a full-scale reading of 15 volt. Find resolution of the converter.

A. 0.00015 volt

B. 0.0015 volt ✓

C. 1.5 volt

D. 0.015 volt



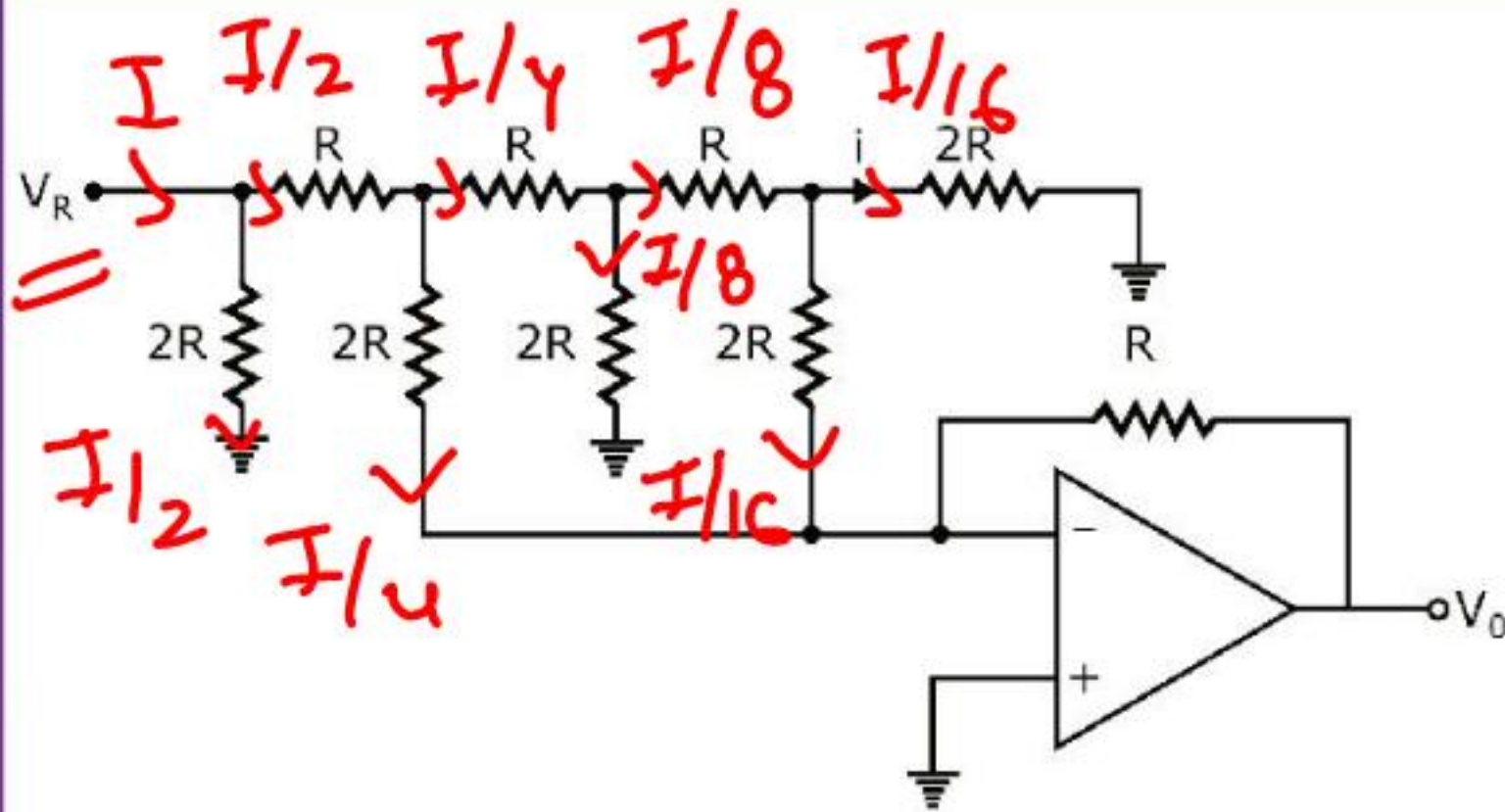
Q. In the digital-to-analog converter circuit shown in the figure below, ~~V_R~~ $V_R = 10V$ and $R = 10 k\Omega$.

The current i is

- A. $31.25 \mu A$
- B. $62.5 \mu A$
- C. $125 \mu A$
- D. $250 \mu A$

$$I = \frac{V_R}{R_{eq}} \\ = \frac{V_R}{R}$$

R-2R Ladder N/w

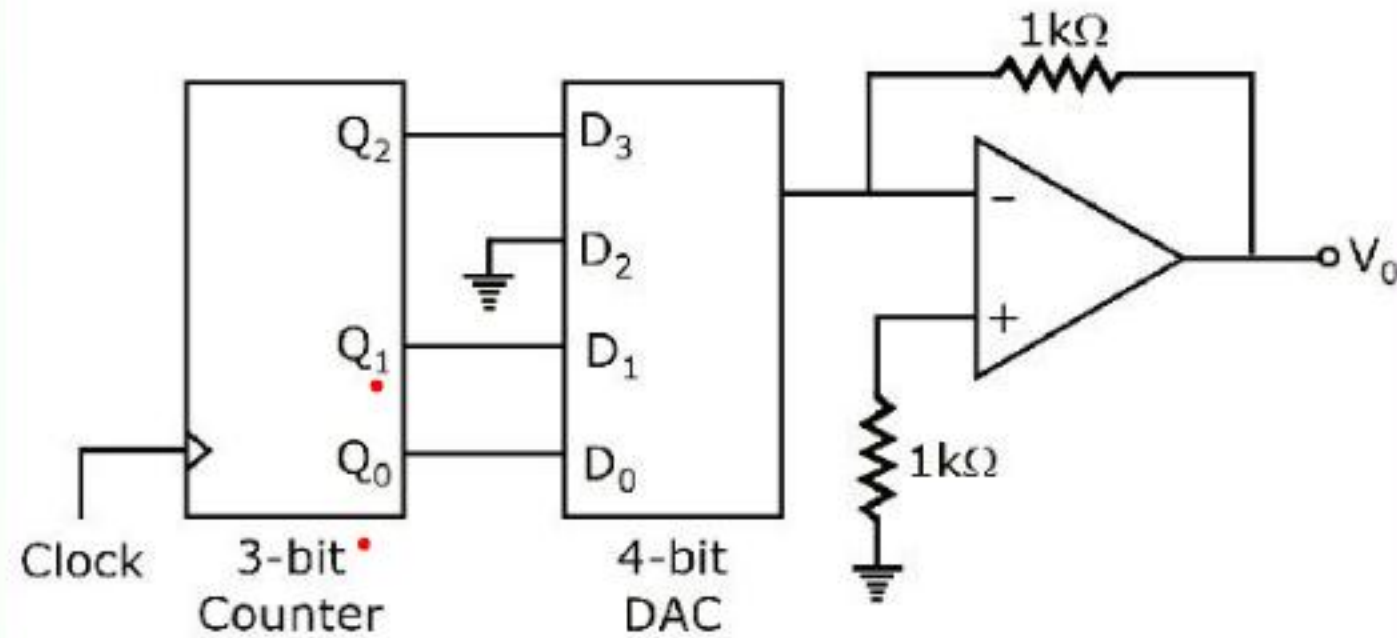


$$i = I/16 = \frac{V}{16R}$$

$$= \frac{10}{16 \times 10}$$

$$= \frac{1}{16} \text{ mA}$$

==

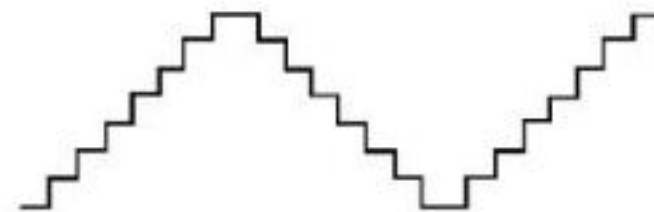


Q. A 4-bit D-to-A converter is connected to a free-running 3-bit UP counter, as shown in the following figure. Which of the following waveforms will be observed at V_0 ?

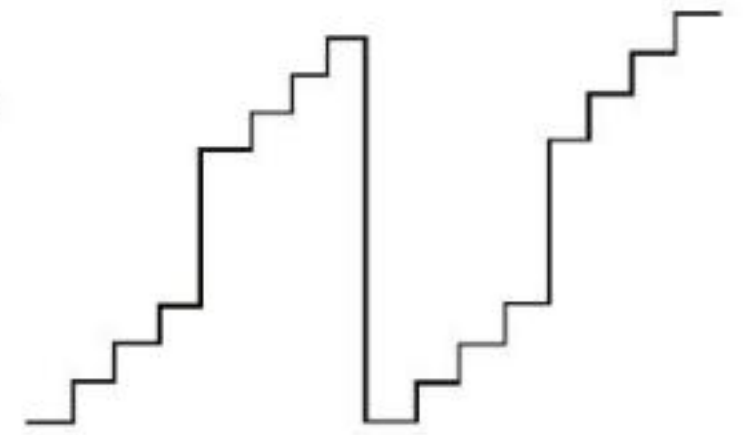
In the figure shown above, the ground has been shown by the symbol

H.W.

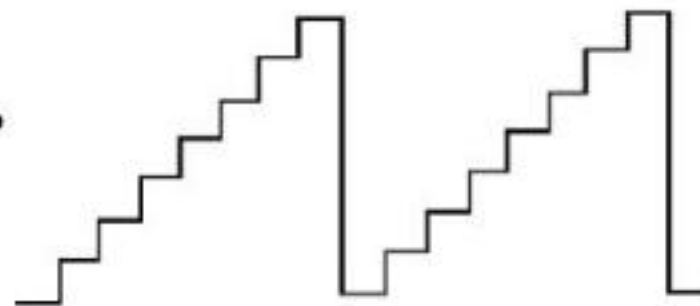
A.



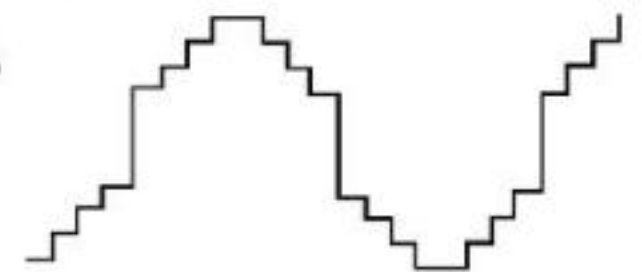
B.

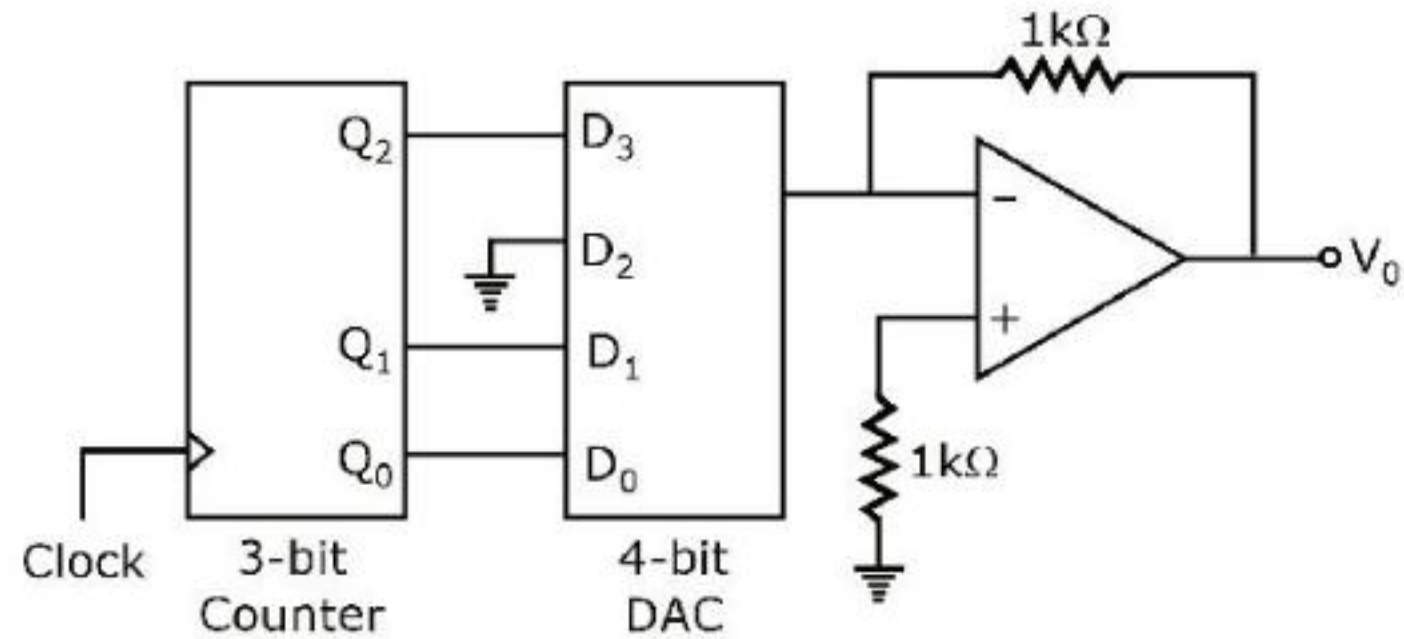


C.



D.





$$2^n - 1$$
$$\underline{\underline{2^8 - 1}}$$

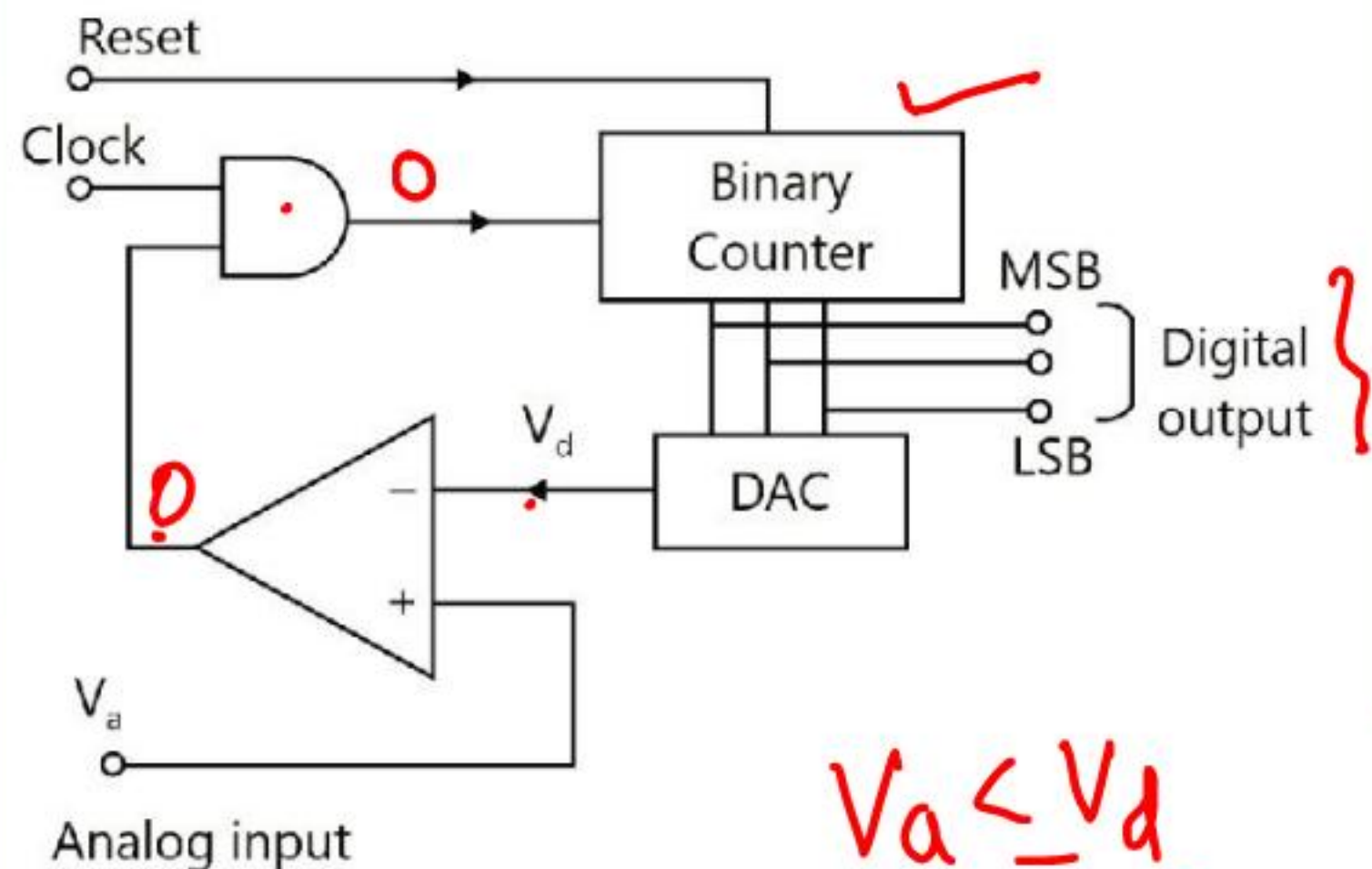
Q. The minimum number of comparators required to build an 8-bit flash ADC is

A. 256

✓ B. 255

C. 512

D. None



Q. At what condition the digital to analog conversion is stopped?

A. $V_a > V_d$

B. $V_a \leq V_d$

C. $V_a \geq V_d$

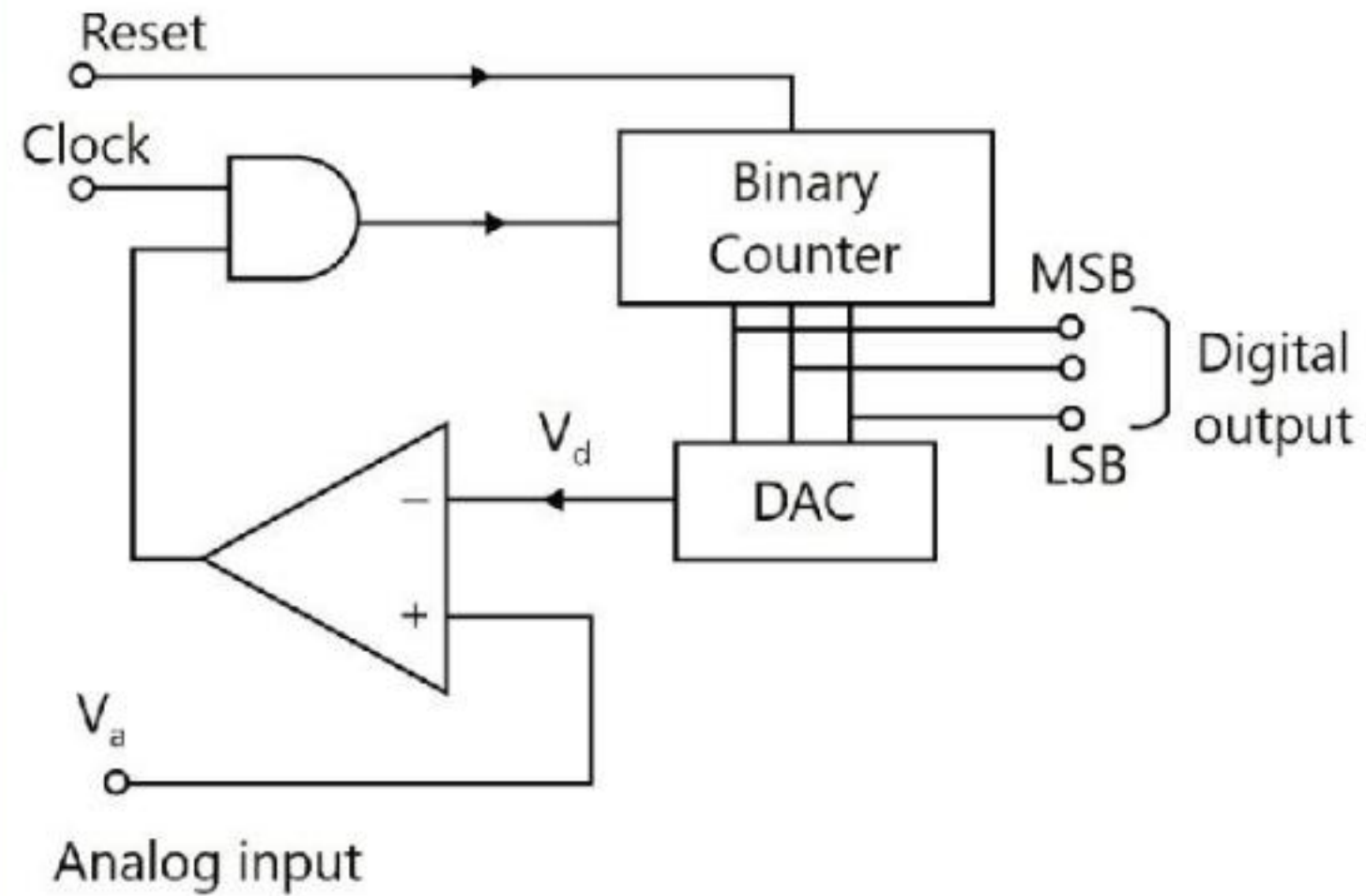
D. $V_a \neq V_d$

Counter Type

ADC

$V_a < V_d$

SAR



NTC

EOC }
SOC }

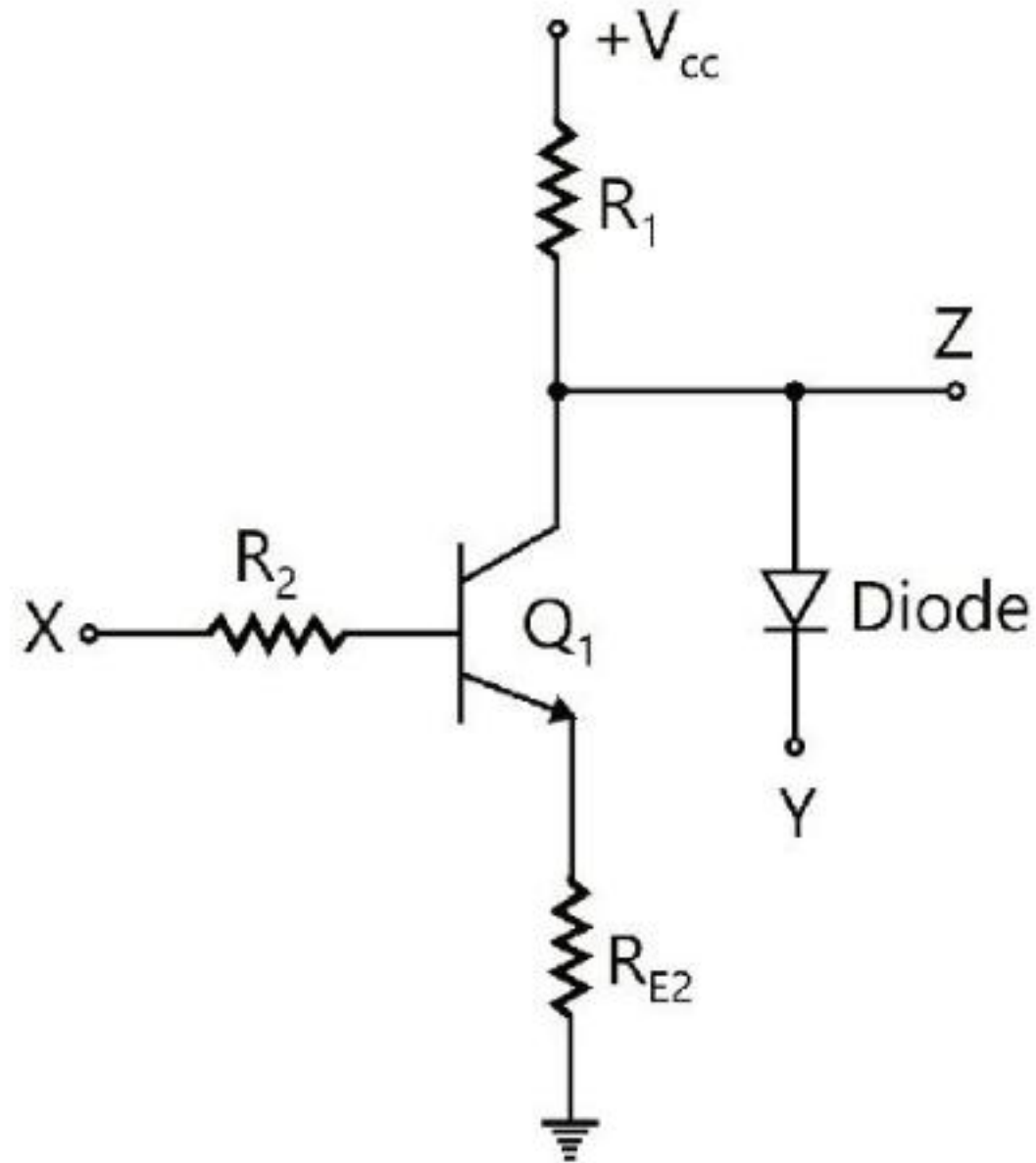
Q. An 8-bit successive approximation Analog to digital converter has full scale reading of 2.55 V and its conversion time for an analog input of 1V is 20 μ sec. The conversion time for a 2V input will be

A. 10 μ s

B. 40 μ s

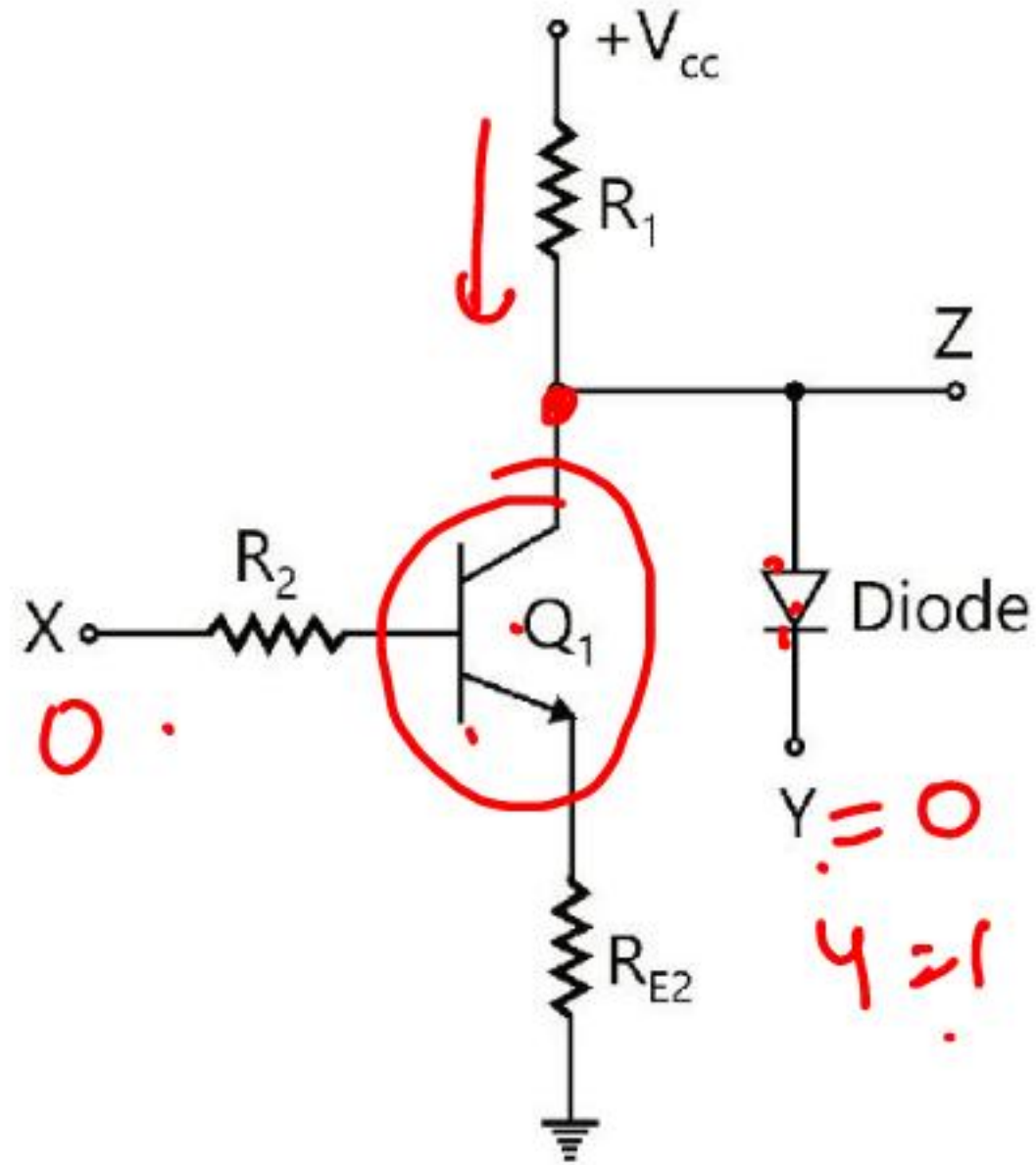
✓ C. 20 μ s

D. 100 μ s



Q. In the circuit shown, Q_1 has negligible collector-to-emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If V_{cc} is +5 V, X and Y are digital signals with 0 V as logic 0 and V_{cc} as logic 1, the Boolean expression for Z is

- A. XY
- B. $X'Y$
- C. XY'
- D. $(XY)'$



X	Y	BJT Mode		Z
0	0	off	on	0
0	1	off	off	1
1	0	on	on	0
1	1	on	off	0

$\overline{X} Y = Z$

mco.

Q. Given that for logic family,

V_{OH} is minimum output high-level voltage.

V_{OL} is the maximum output-low-level voltage.

V_{IH} is the minimum acceptable input high-level voltage and

V_{IL} is the maximum acceptable input low-level voltage,

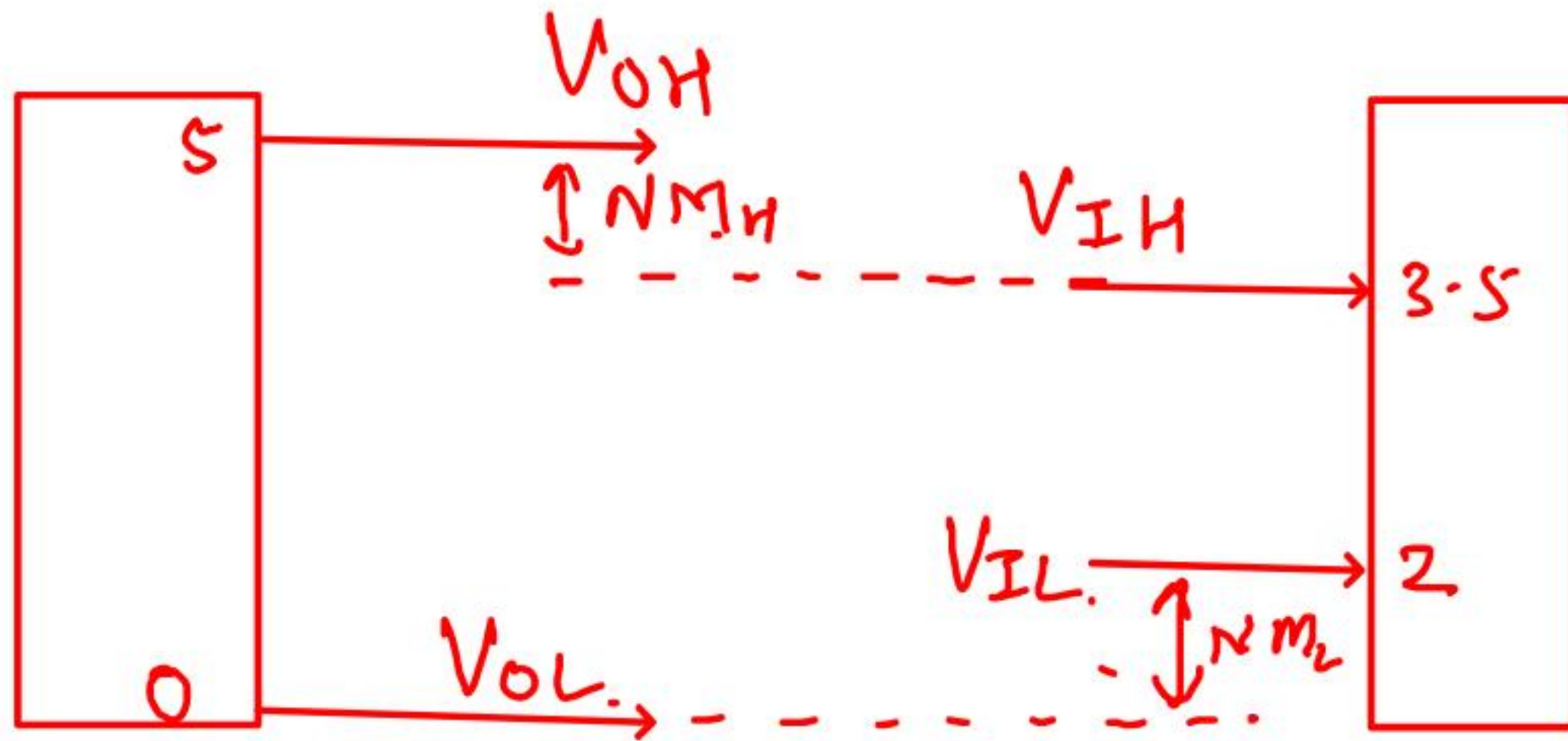
The correct relationship is

A. $V_{IH} > V_{OH} > V_{IL} > V_{OL}$

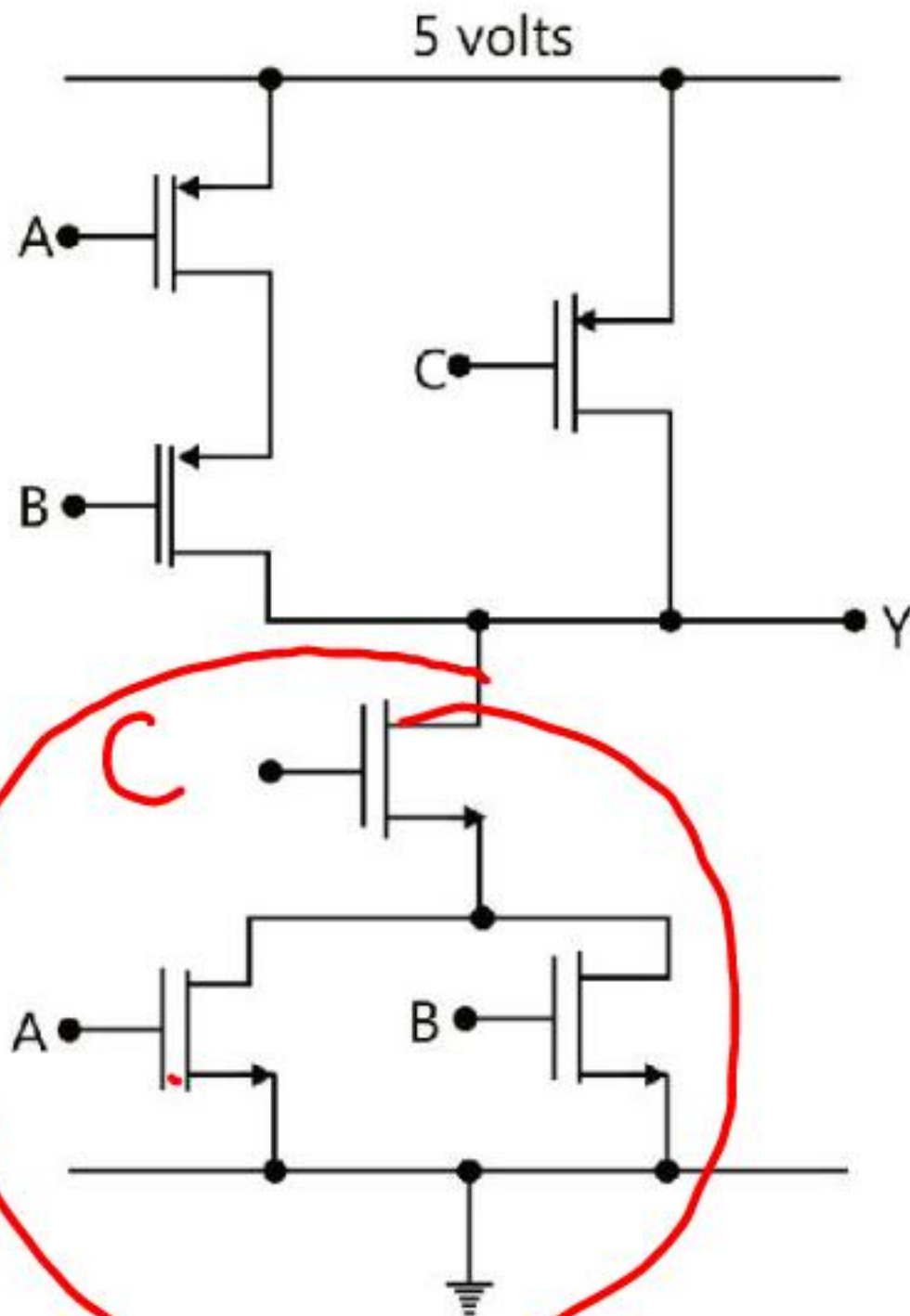
☒ B. $V_{OH} > V_{IH} > V_{IL} > V_{OL}$

C. $V_{IH} > V_{OH} > V_{OL} > V_{IL}$

D. $V_{OH} > V_{IH} > V_{OL} > V_{IL}$



$$NM = 1.5$$



Q. In the circuit shown

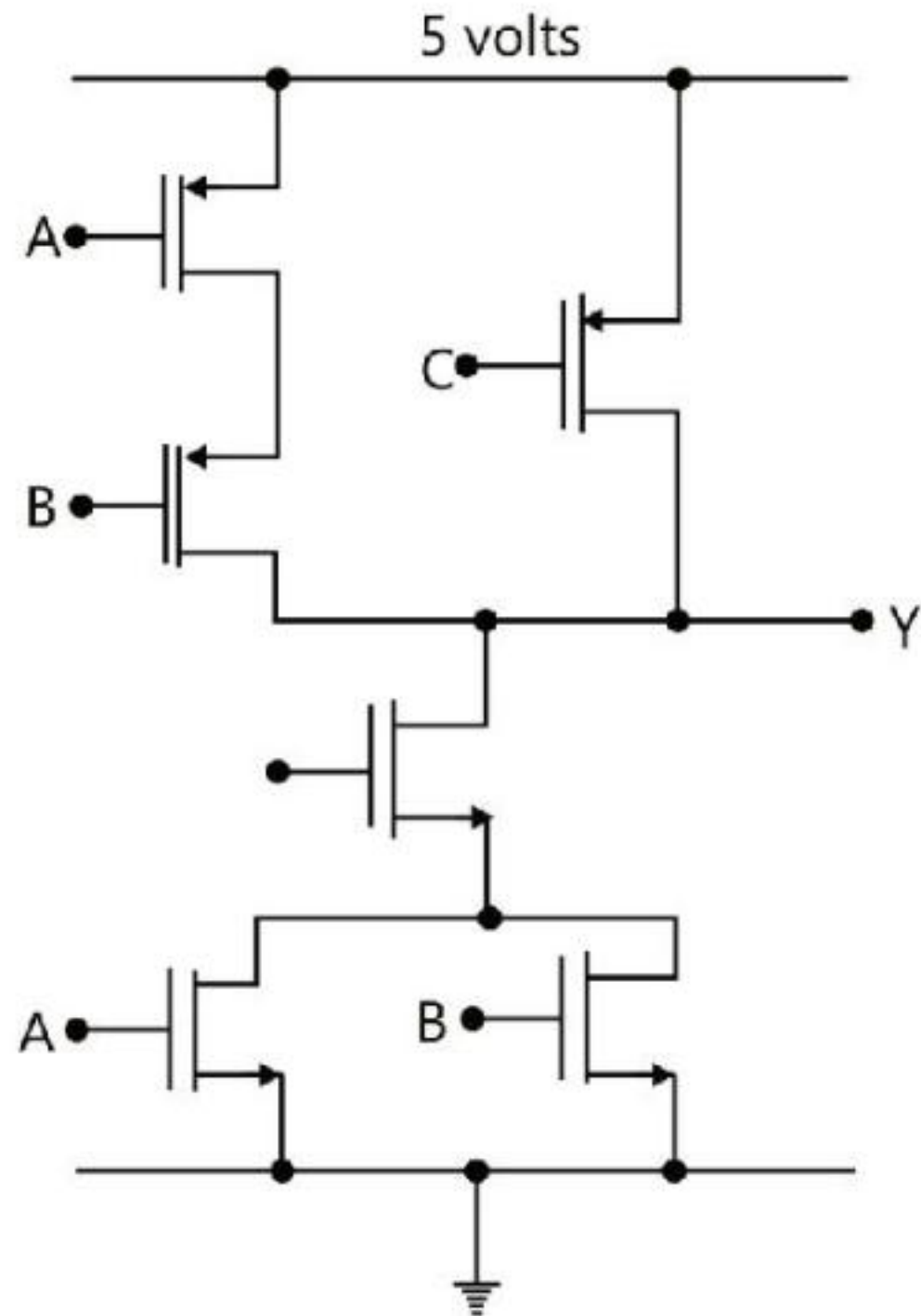
- ✓ A. $Y = \bar{A}\bar{B} + \bar{C}$
- B. $Y(A + B)C$
- C. $Y = (\bar{A} + \bar{B})\bar{C}$
- D. $Y = \bar{A}\bar{B} + C$

$$Y = \overline{(A + B)C}$$

$$= \overline{A + B} + \bar{C}$$

$$= \bar{A}\bar{B} + \bar{C}$$

$$C(A + B)$$



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